

Authored Grade-9 Force and Laws of Motion spine covering balanced/unbalanced for: Transfer

A crate rests on a rough horizontal floor. It is pulled by a force P at an upward angle θ . The coefficients μ_s and μ_k are known, but the statement does not say whether the crate slips. For a specified P , determine the crate's acceleration or justify that it remains at rest.

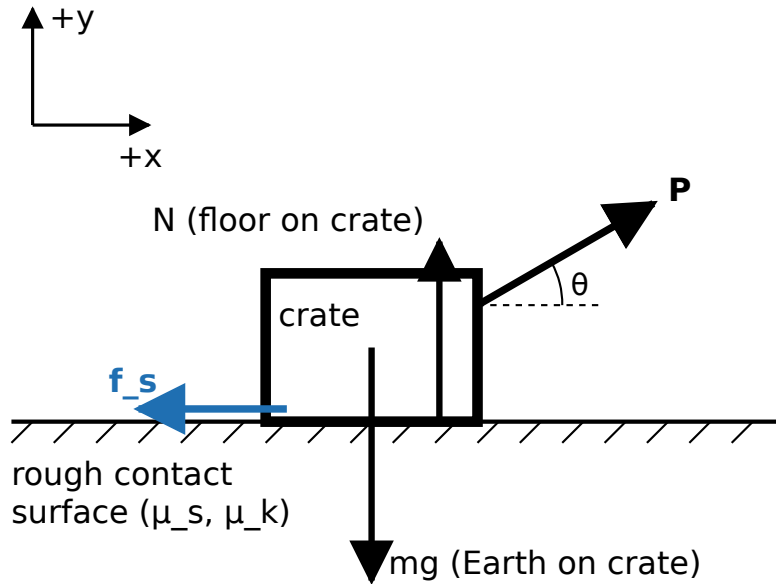
Builds on

- [A small block of mass \$m\$ rests on a larger block of mass \$M\$. The floor under the larger...](#)

Conditions

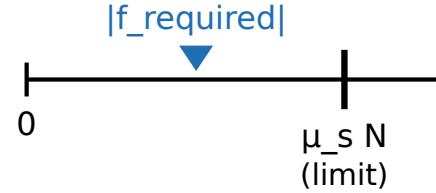
- Resolve the applied force into horizontal and vertical components.
- The crate remains in contact with the floor.
- Do not assume a friction equality before deciding the contact state.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



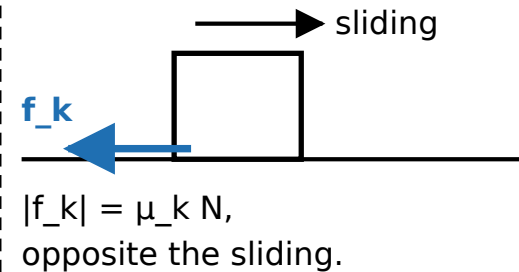
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

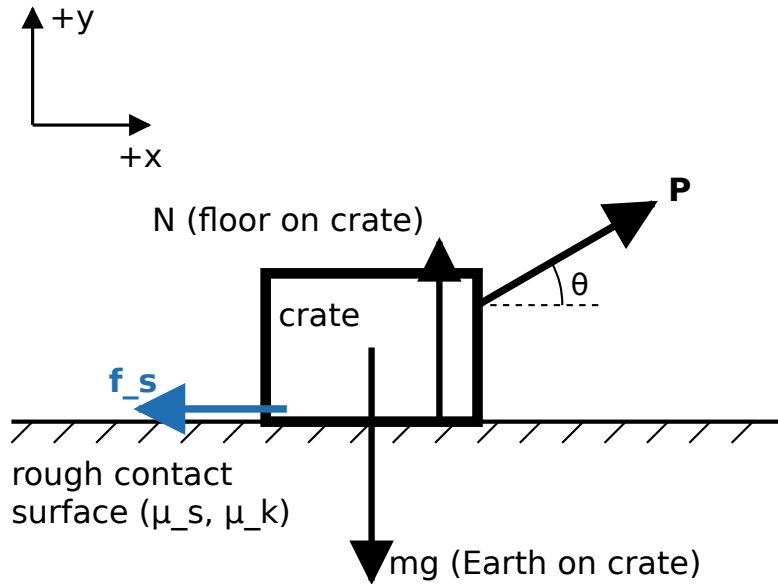
What changed

The learner must infer the contact state and choose the static-feasibility or kinetic-friction model instead of being told that the bodies are at impending slip or already sliding.

What stayed valid

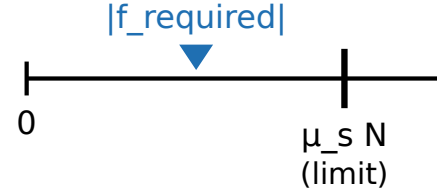
The normal force from the actual force balance, the provisional no-slip requirement and its comparison with $\mu_s N$ all carry over; what changes is that the contact state is not given and must be decided.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



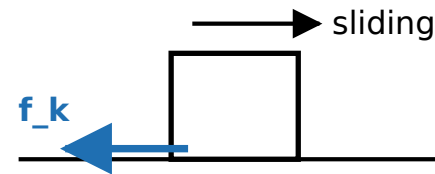
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

First find $N = mg - P \sin(\theta)$. Under a provisional no-slip state, the friction required is $f_{\text{required}} = P \cos(\theta)$. If $|f_{\text{required}}| \leq \mu_s N$, the crate can remain at rest and static friction equals the required value. If that inequality fails, sliding occurs; then use kinetic friction $\mu_k N$ opposite the sliding and apply Newton II to find acceleration.

1. The upward component of P changes the normal reaction, so N is not automatically mg .
2. The coefficient data do not identify the contact state; test the static state first.
3. Compare the no-slip friction requirement with the static limit.
4. Only after the static state is impossible is the kinetic-friction model selected for the acceleration calculation.

Rubric

- Computes the actual normal reaction before any friction limit.
- Tests the no-slip friction requirement against $\mu_s N$ before choosing a kinetic model.
- Uses kinetic friction only after static feasibility fails.

Independent check

At $P=0$, the method must allow friction to be zero rather than $\mu_s mg$.

Repair

Revisit: Assume no relative sliding provisionally and solve the tangential Newton-II equation for the static friction value f_{required} that would make that motion possible.

Lineage check

The normal force from the actual force balance, the provisional no-slip requirement and its comparison with $\mu_s N$ all carry over; what changes is that the contact state is not given and must be decided.

The learner must infer the contact state and choose the static-feasibility or kinetic-friction model instead of being told that the bodies are at impending slip or already sliding.

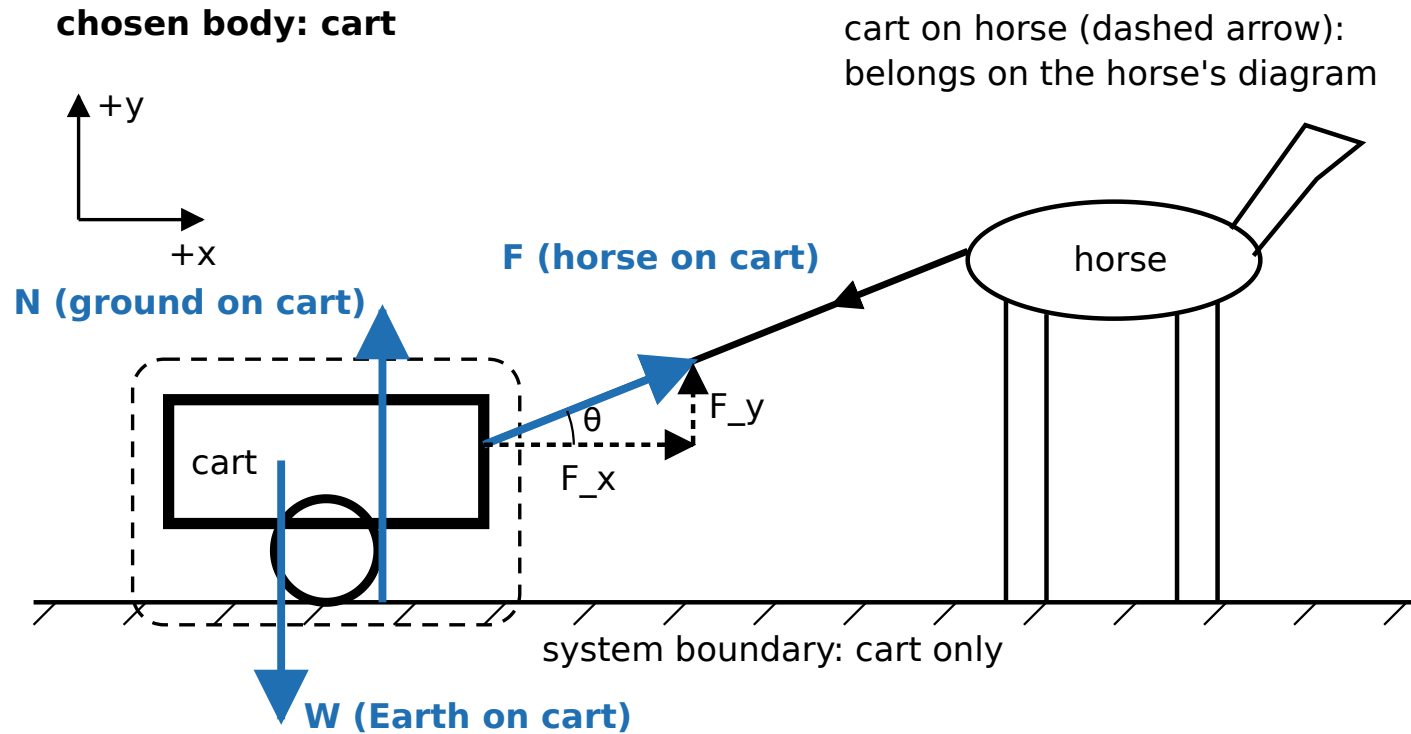
Three identical carts A, B and C move in a line on a smooth track. A pushes B, B pushes C, and contact is maintained while an external force F is applied only to A. Without being told which system to choose, find the acceleration of the three-cart train and the contact force that B exerts on C.

Builds on

- [Two blocks A and B of masses \$m_A\$ and \$m_B\$ remain in contact on a smooth horizontal...](#)

Conditions

- The carts remain in contact.
- The track is frictionless.
- Take the direction of F as positive.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

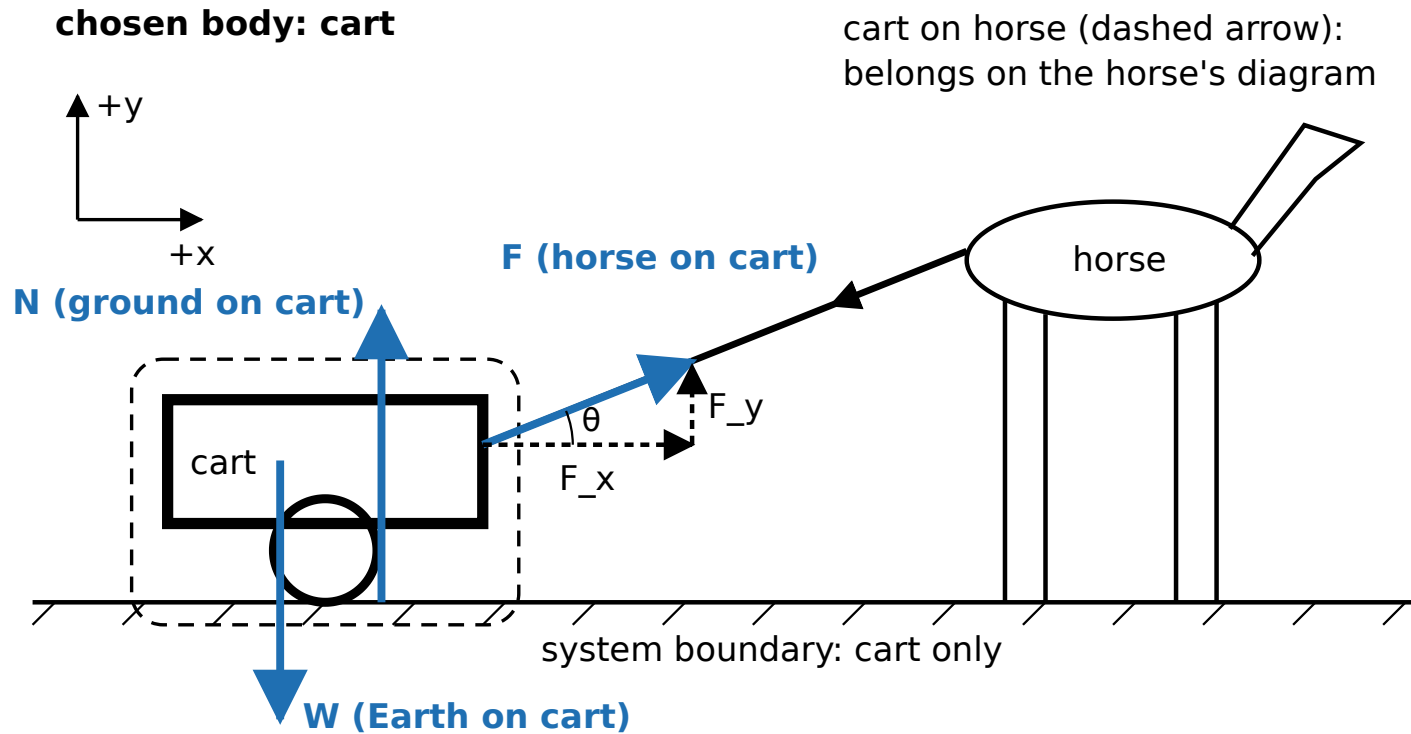
Review and solution

What changed

The learner must choose and change system boundaries across a multi-body solution, rather than being handed separate-body versus whole-system equations.

What stayed valid

Maintained contact still gives one shared acceleration, and a contact force still belongs only in the equation of the body it acts on; what changes is that you choose which system to isolate.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

Use A+B+C as one system for the acceleration: $a = F/(3m)$. Then isolate C to find the requested contact force: $N_{BC} = m a = F/3$. The B-on-C force is internal only to the three-cart system; it is external to cart C and must remain in C's equation.

1. Maintained contact gives the carts one common acceleration along the track.
2. For the acceleration, the whole three-cart system removes internal contact forces from the summed equation.
3. For the requested B-on-C contact force, isolate C because that interaction is external to C.
4. Newton II on C gives $N_{BC} = mF/(3m) = F/3$.

Rubric

- Chooses the whole train for the acceleration calculation.
- Switches to cart C for the requested contact force.
- Keeps the same common acceleration across both equations.

Independent check

Cancelling B-on-C while solving cart C alone would leave C with no horizontal force but nonzero acceleration, which is inconsistent.

Repair

Revisit: Match the chosen system boundary to the requested quantity: use the combined constrained system when internal forces are irrelevant to the shared acceleration, then isolate the body or subsystem that makes a requested interaction force external.

Lineage check

Maintained contact still gives one shared acceleration, and a contact force still belongs only in the equation of the body it acts on; what changes is that you choose which system to isolate.

The learner must choose and change system boundaries across a multi-body solution, rather than being handed separate-body versus whole-system equations.

Two masses are connected by one rope over a fixed pulley. The pulley axle is explicitly stated to have appreciable friction. A proposed solution writes the same tension T on both sides of the pulley before doing any calculation. Decide whether that step is justified by the Grade9V3 ideal-string model and explain what can still be said without introducing a new pulley model.

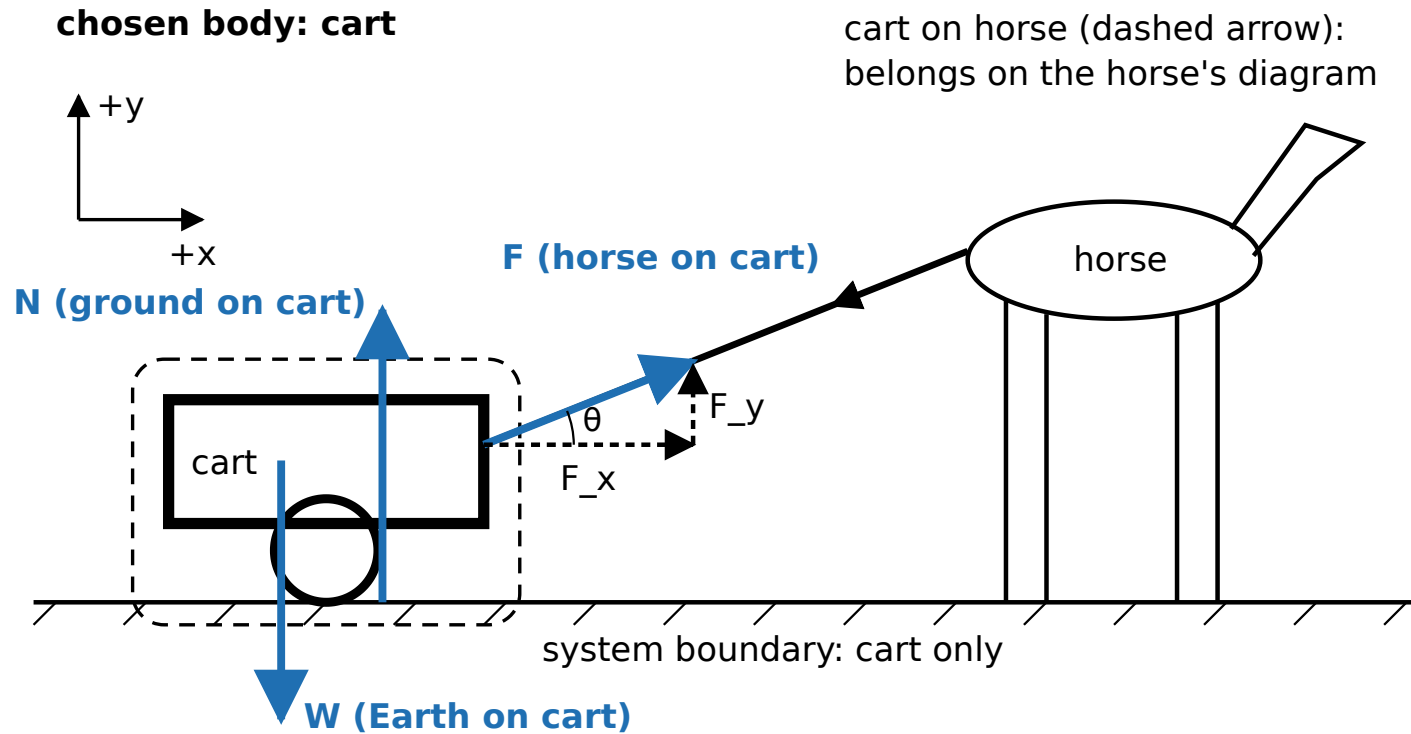
Builds on

- [Two carts A and B of masses \$m_A\$ and \$m_B\$ are connected by one taut massless...](#)

Conditions

- The rope remains taut.
- The pulley axle is not frictionless.

- No quantitative nonideal-pulley relation is supplied.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

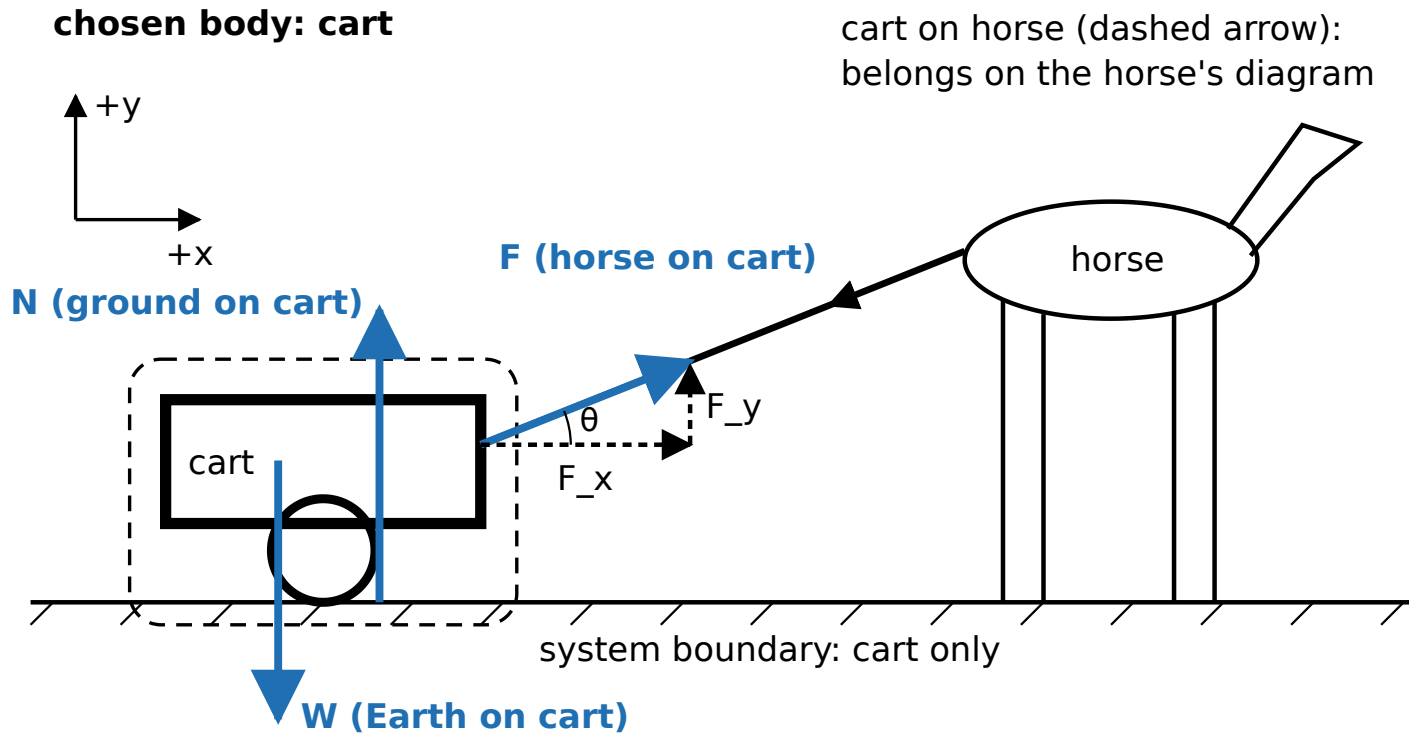
Review and solution

What changed

The learner must decide whether the equal-tension ideal-string model is valid after the problem explicitly breaks an ideal-pulley assumption.

What stayed valid

Each body still has its own tension force along its own string segment; what changes is whether the ideal assumptions that allowed one T still hold.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

The one-common-T step is not justified by the bounded ideal-string/redirection model because a required ideal assumption has been broken. Each mass still has a rope tension force on its own free-body diagram, but this capability must not assert equal magnitudes across the nonideal pulley. A quantitative solution would require an additional model that is outside this slice.

1. Tension forces still belong on the bodies they act on.
2. Uniform tension across a redirection was licensed only after declaring a frictionless, massless ideal pulley/string model.
3. Appreciable axle friction breaks that declared model condition.
4. Therefore equal tension cannot be imported; the correct bounded response is to stop rather than manufacture a nonideal relation.

Rubric

- Rejects one common tension because the ideal redirection condition is broken.
- Keeps a tension force on each mass's own FBD.
- Refuses to invent a quantitative nonideal-pulley relation.

Independent check

If the statement is changed back to a frictionless massless fixed pulley with a massless taut string, the equal-tension relation becomes licensed again.

Repair

Revisit: Check whether the problem introduces rope mass, elasticity, pulley friction, or rotational inertia; if it does, stop using equal tension unless a different model justifies it.

Lineage check

Each body still has its own tension force along its own string segment; what changes is whether the ideal assumptions that allowed one T still hold.

The learner must decide whether the equal-tension ideal-string model is valid after the problem explicitly breaks an ideal-pulley assumption.

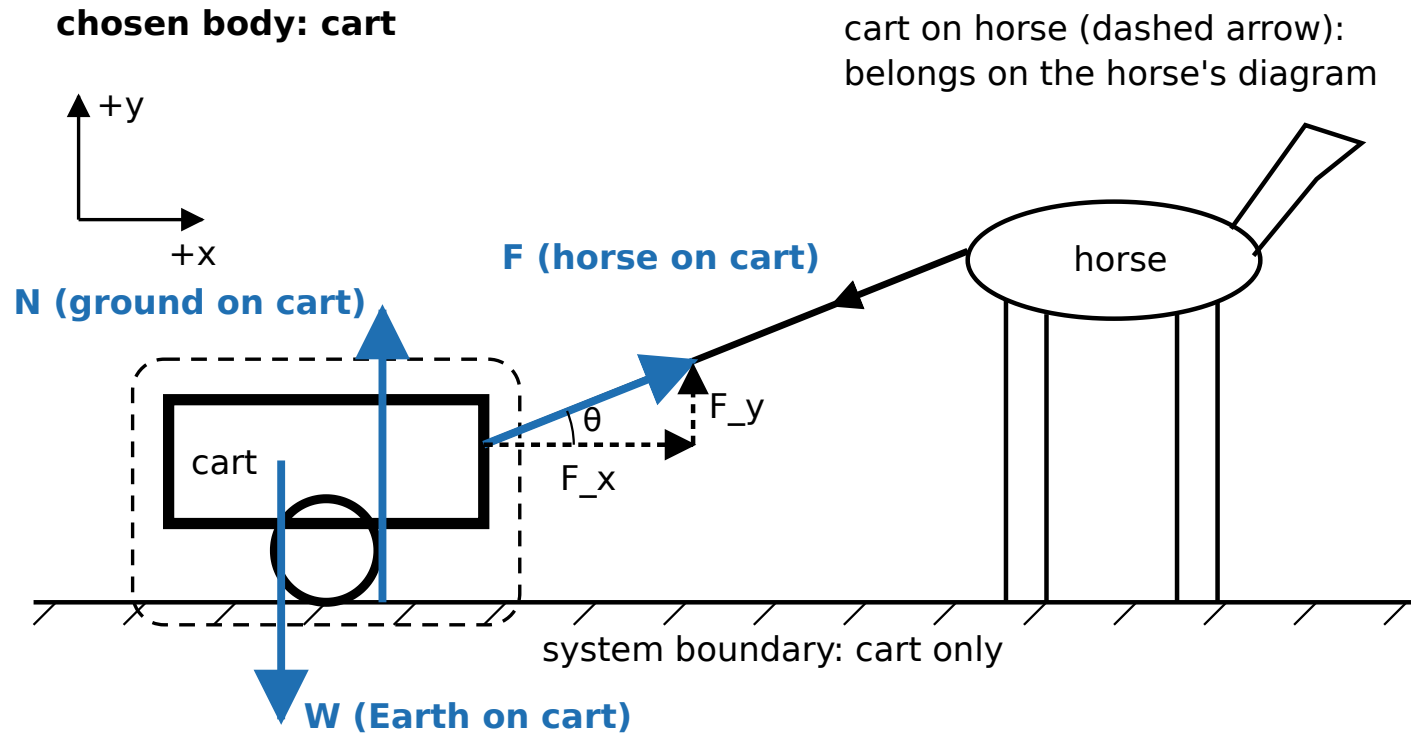
A diagram is replaced by this verbal description: one taut inextensible string passes over one fixed pulley, with mass A on the left end and mass B on the right end. Coordinates x_A and x_B are both defined positive upward. Before writing any Newton-II equations, construct the string-length relation and derive the signed acceleration relation in these coordinates.

Builds on

- [One taut inextensible string passes over one fixed pulley. Let \$y_A\$ and \$y_B\$ be the two...](#)

Conditions

- The pulley is fixed.
- Only the two vertical straight string segments change length.
- Both coordinates are positive upward, toward the pulley.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

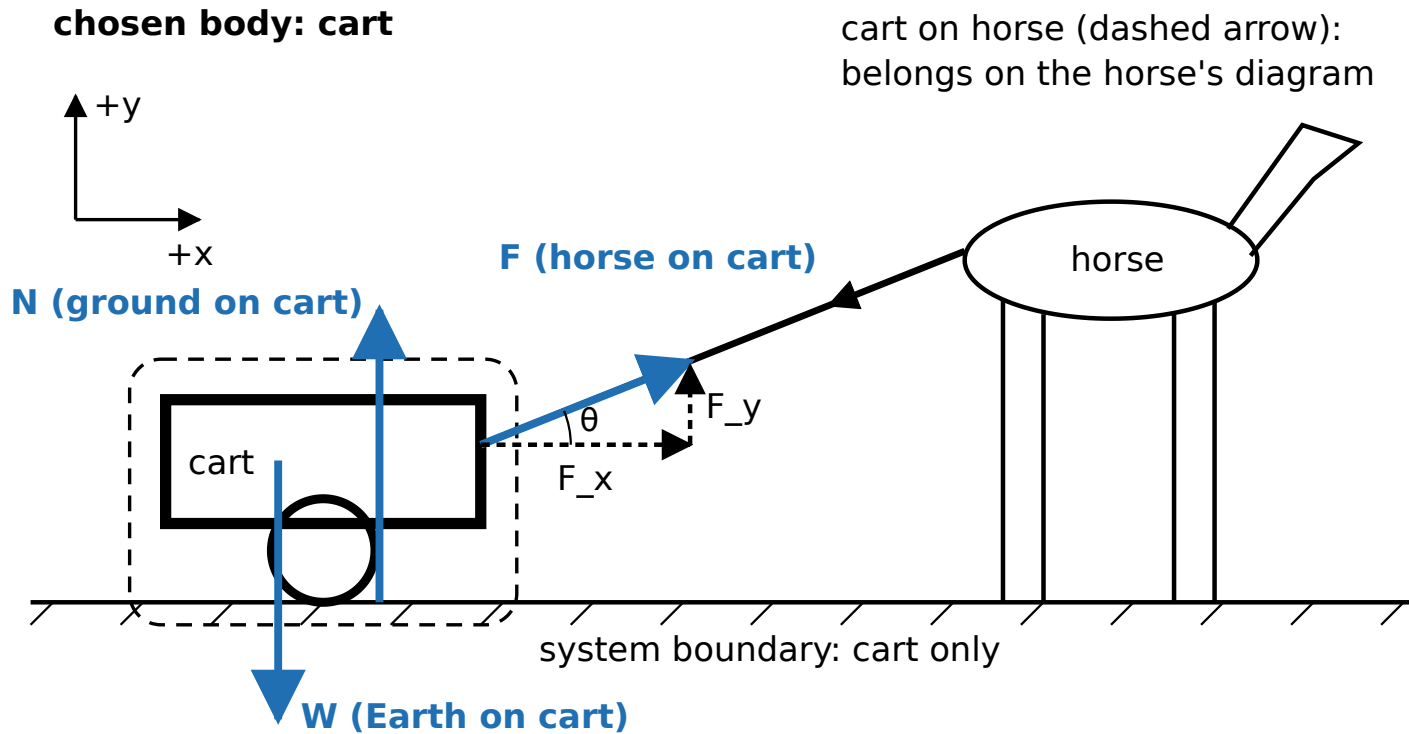
Review and solution

What changed

The learner must translate a verbal fixed-pulley geometry with a different coordinate convention into the string-length equation before obtaining the acceleration constraint.

What stayed valid

One fixed pulley and one inextensible string still fix the total changing length; what changes is the coordinate direction, and that the geometry arrives in words.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

Because the changing segment lengths decrease when x_A or x_B increases, the variable string length can be written $(C_A - x_A) + (C_B - x_B) = \text{constant}$. Hence $x_A + x_B = \text{constant}$, so $v_A + v_B = 0$ and $a_A + a_B = 0$. Thus if A accelerates upward, B accelerates downward with equal magnitude.

1. Translate the verbal geometry into the two changing string-segment lengths.
2. The chosen coordinates are upward, so each segment length is a constant offset minus its coordinate.
3. Inextensibility makes the sum of the segment lengths constant.
4. Differentiating the equivalent $x_A + x_B = \text{constant}$ relation gives the signed velocity and acceleration constraints.

Rubric

- Builds the variable string length from the verbal geometry before writing an acceleration relation.
- Uses the declared upward coordinate signs consistently.
- Derives $a_A + a_B = 0$ from the length relation.

Independent check

If both masses accelerated upward in these coordinates, both vertical segments would shorten and the fixed total string length would be violated.

Repair

[Revisit: Write the variable part of the string length as \$y_A + y_B\$ and set its total equal to a constant.](#)

Lineage check

One fixed pulley and one inextensible string still fix the total changing length; what changes is the coordinate direction, and that the geometry arrives in words.

The learner must translate a verbal fixed-pulley geometry with a different coordinate convention into the string-length equation before obtaining the acceleration constraint.

A small bob hangs from the roof of a car that accelerates horizontally to the right. A student wants the bob to be at rest in the chosen coordinates while finding the string direction. Decide which observer description makes that possible, state the extra modelling term required there, and give an independent inertial-frame alternative without solving for a numerical angle.

Builds on

- [A bus accelerates to the right on a straight road. A loose object on a smooth...](#)

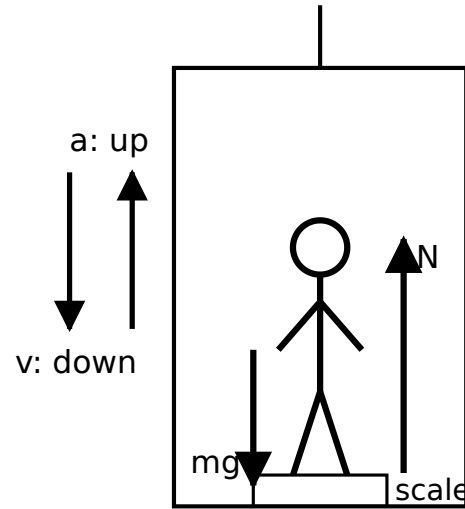
Conditions

- The road is horizontal.
- The car acceleration is constant and nonzero.

- Air resistance is neglected.

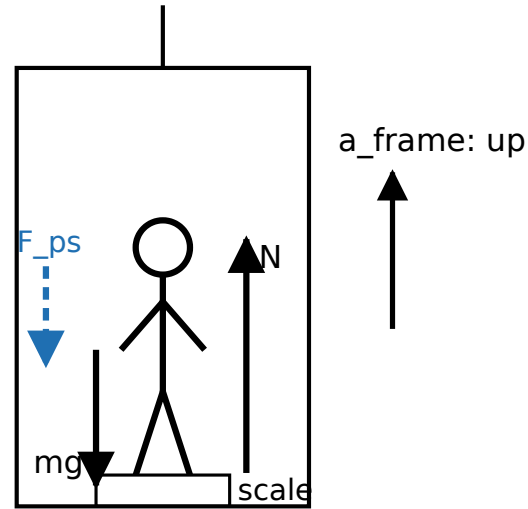
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

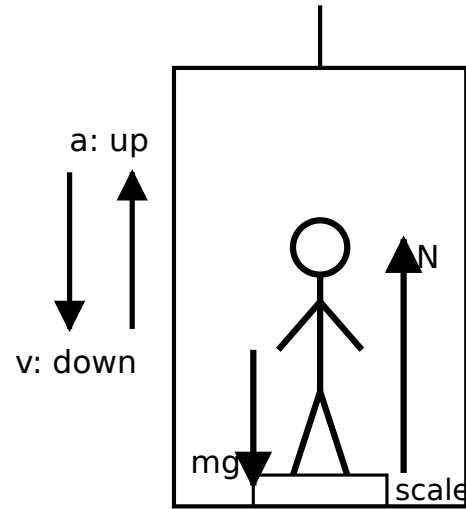
The learner must choose between an inertial roadside analysis and an accelerating-car analysis from the task's representation goal, instead of being told which frame to use.

What stayed valid

The physical forces on the bob, gravity and the string's tension, are the same for every observer, and both descriptions must predict one string direction; what changes is that you choose which observer to use.

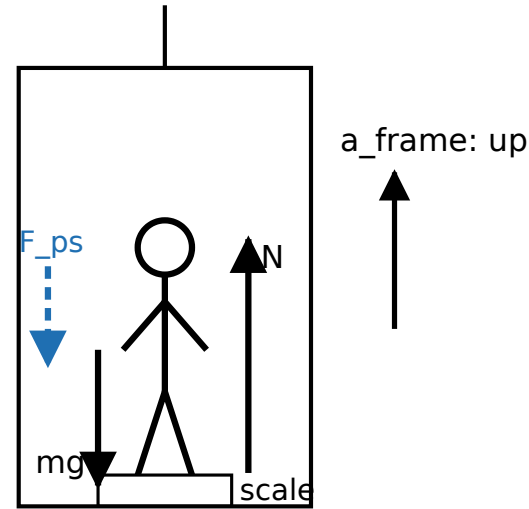
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 5

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Answer

To keep the bob at rest in the chosen coordinates, use the accelerating car frame and include a pseudo-force $m a_{\text{car}}$ to the left in addition to the physical forces. Alternatively, use the roadside inertial frame with only physical forces and give the bob the same rightward acceleration as the car. Both descriptions must predict the same string direction.

1. The requirement that the bob be at rest in the coordinates points to the car frame.
2. Because that observer frame accelerates, Newton's second law there needs the backward pseudo-force convention.
3. In the roadside inertial frame no pseudo-force is added; the bob has the car's horizontal acceleration.
4. The two descriptions are alternative models of the same physical situation and must agree on the observable string direction.

Rubric

- Chooses the car frame only because the task requests the bob to be stationary in the chosen coordinates.
- Adds the pseudo-force specifically because the car frame accelerates.
- States a valid roadside alternative with physical forces only and horizontal bob acceleration.

Independent check

If the car acceleration becomes zero, the pseudo-force term vanishes and both descriptions reduce to the ordinary hanging-bob equilibrium.

Repair

[Revisit: Add a backward pseudo-force only in the accelerating-observer description and compare it with the roadside account.](#)

Lineage check

The physical forces on the bob, gravity and the string's tension, are the same for every observer, and both descriptions must predict one string direction; what changes is that you choose which observer to use.

The learner must choose between an inertial roadside analysis and an accelerating-car analysis from the task's representation goal, instead of being told which frame to use.

A runner accelerates forward without the shoe slipping on the ground. Determine the direction of the static friction force exerted by the ground on the shoe, and justify it from the relative slip tendency at the contact.

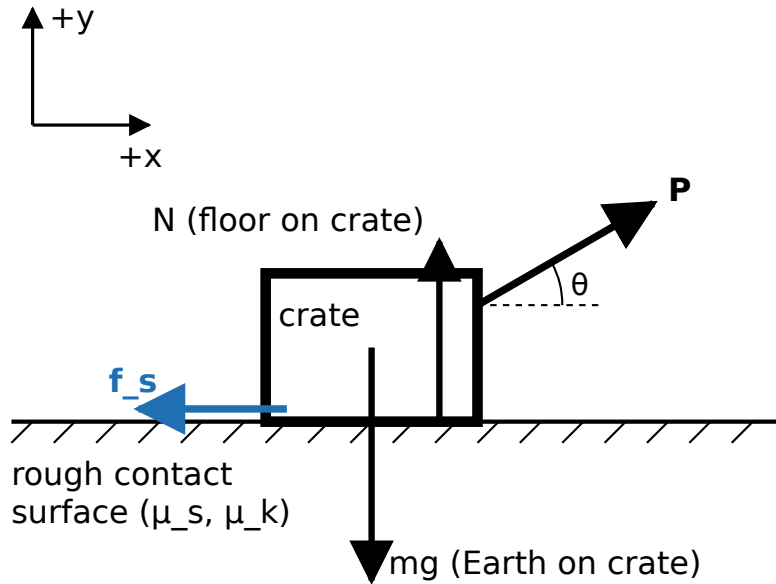
Builds on

- [A box is placed on a conveyor belt moving to the right faster than the box. Before the...](#)

Conditions

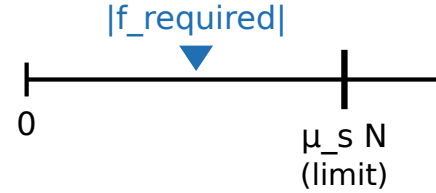
- The shoe does not slip on the ground during the instant considered.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



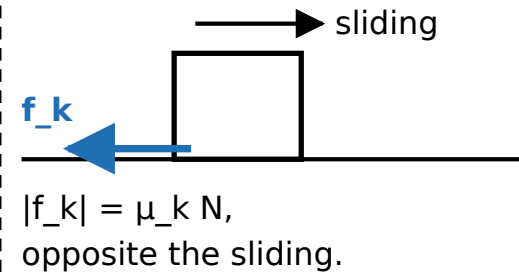
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$: Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

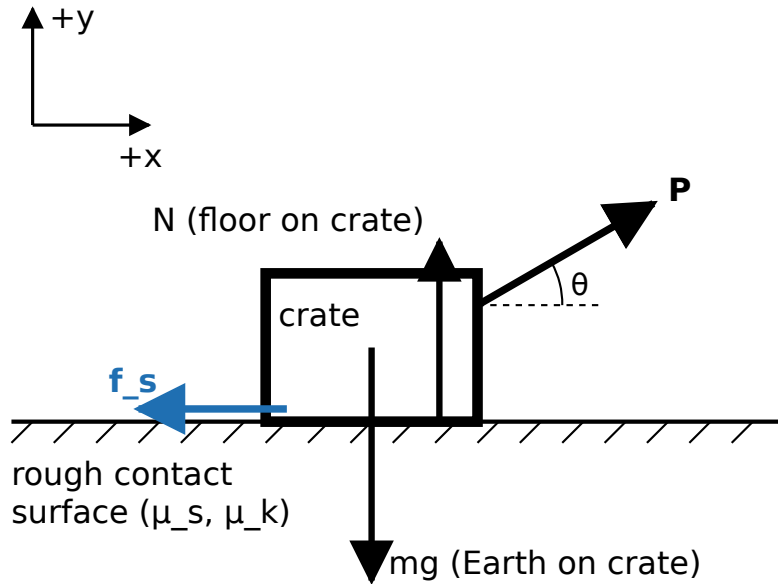
What changed

The learner must translate the same relative-slip rule from a belt-box contact to a shoe-ground interaction without using the object's ground-frame motion as the sign rule.

What stayed valid

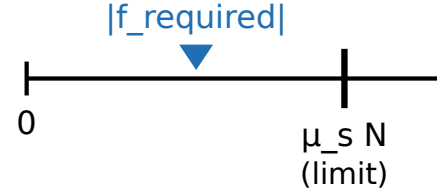
Friction still opposes the slip tendency at the contact, whatever the object's overall velocity; only the contact changes, from a belt and a box to a shoe and the ground.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



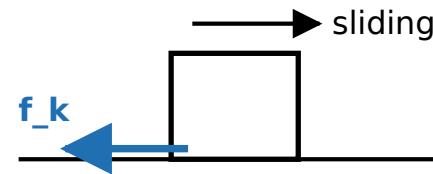
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

The ground's static friction on the shoe points forward.

1. Without sufficient friction, the runner's foot would tend to slip backward relative to the ground while pushing against it.
2. Static friction opposes that relative slip tendency.
3. Therefore the ground exerts forward friction on the shoe, helping accelerate the runner forward.

Rubric

- Identifies the backward relative slip tendency of the shoe at the contact if friction were absent.
- Places the ground's friction force on the shoe forward.
- Does not invoke 'friction always opposes motion' as the justification.

Independent check

The friction direction may be the same as the runner's ground-frame velocity; the defining comparison is the contact-level relative slip tendency.

Repair

Revisit: Imagine the relative sliding or tendency to slide if friction were absent, then point friction opposite that relative tendency.

Lineage check

Friction still opposes the slip tendency at the contact, whatever the object's overall velocity; only the contact changes, from a belt and a box to a shoe and the ground.

The learner must translate the same relative-slip rule from a belt-box contact to a shoe-ground interaction without using the object's ground-frame motion as the sign rule.

A rough horizontal platform and a crate on it are already moving together at constant velocity. At the instant considered there is no other horizontal force on the crate. Must a nonzero friction force act on the crate merely because the surfaces are rough?

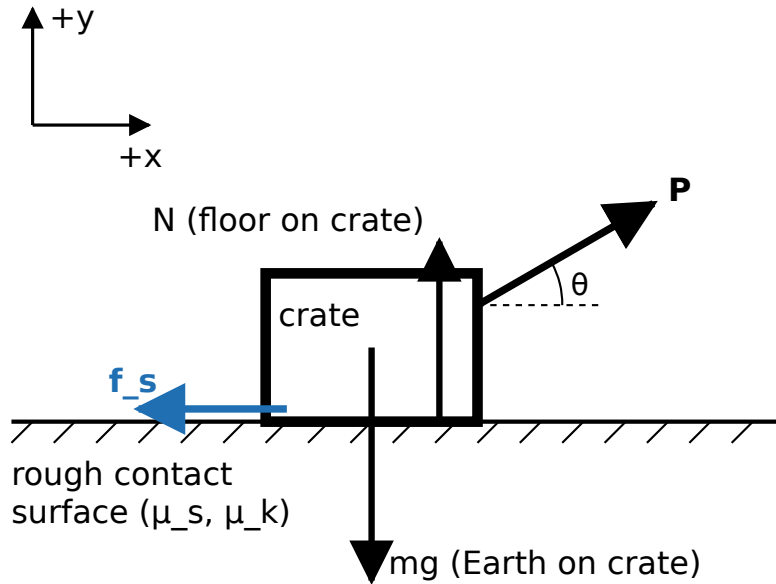
Builds on

- [A crate rests on a rough horizontal floor. No horizontal force acts on it, and there...](#)

Conditions

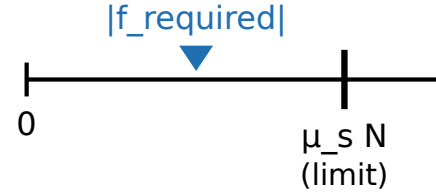
- The crate has no horizontal acceleration.
- The crate and platform have no relative sliding or tendency to separate in speed at the instant considered.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



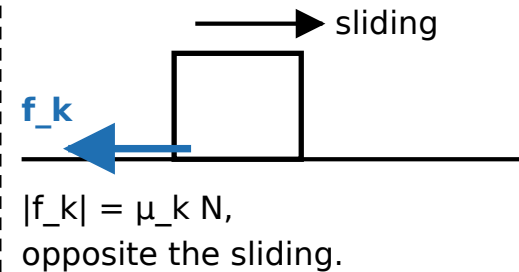
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

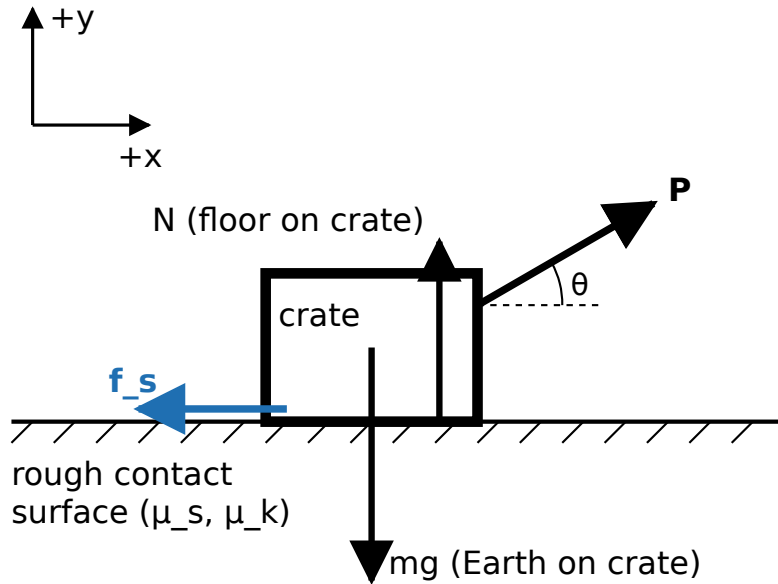
What changed

The learner must decide not to invoke a friction force despite rough moving contact; the contact model is selected from slip tendency, not from the presence of motion or a coefficient.

What stayed valid

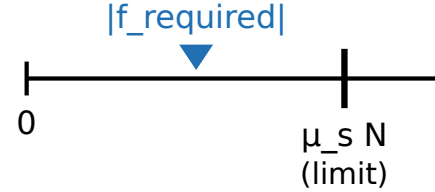
Friction still appears only where the contact has a slip tendency to oppose; what changes is that both surfaces are moving together.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



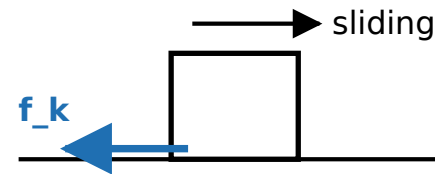
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

No. The friction force can be zero.

1. If friction vanished at this instant, the crate would continue with its current velocity because no other horizontal force changes it.
2. The platform is also moving at that same constant velocity, so no relative slip tendency is created.
3. Therefore no tangential contact force is required and static friction may be zero.

Rubric

- Tests the contact state by imagining friction removed.
- Concludes that no relative slip tendency appears in the stated constant-velocity state.
- Allows $f=0$ despite a rough contact.

Independent check

Roughness sets the range of friction the contact can provide; it does not force the contact to use a nonzero value.

Repair

Revisit: Decide whether any tangential friction is required at all; allow friction to be zero when the contact has neither relative sliding nor a tendency to slide.

Lineage check

Friction still appears only where the contact has a slip tendency to oppose; what changes is that both surfaces are moving together.

The learner must decide not to invoke a friction force despite rough moving contact; the contact model is selected from slip tendency, not from the presence of motion or a coefficient.

A block of mass m is pushed on a rough horizontal floor by a force P directed at an angle θ below the horizontal. The coefficient of static friction is μ_s . Derive the largest P for which the block can remain at rest.

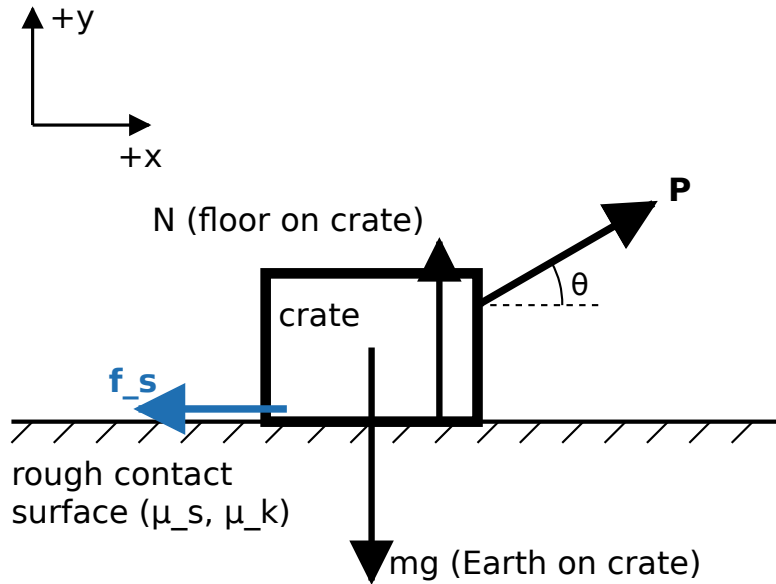
Builds on

- [A 5 kg block rests on a rough horizontal floor. A 12 N horizontal pull is applied, but...](#)

Conditions

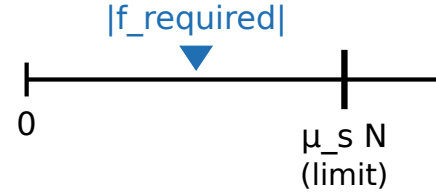
- $0 < \theta < 90$ degrees.
- The block remains in contact with the floor.
- Assume $\cos(\theta) - \mu_s \sin(\theta) > 0$.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



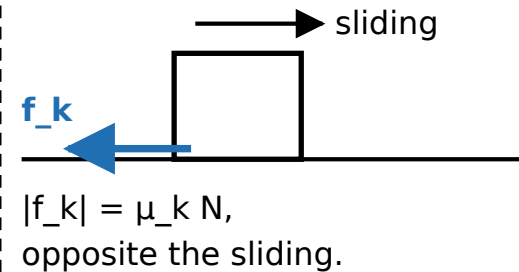
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

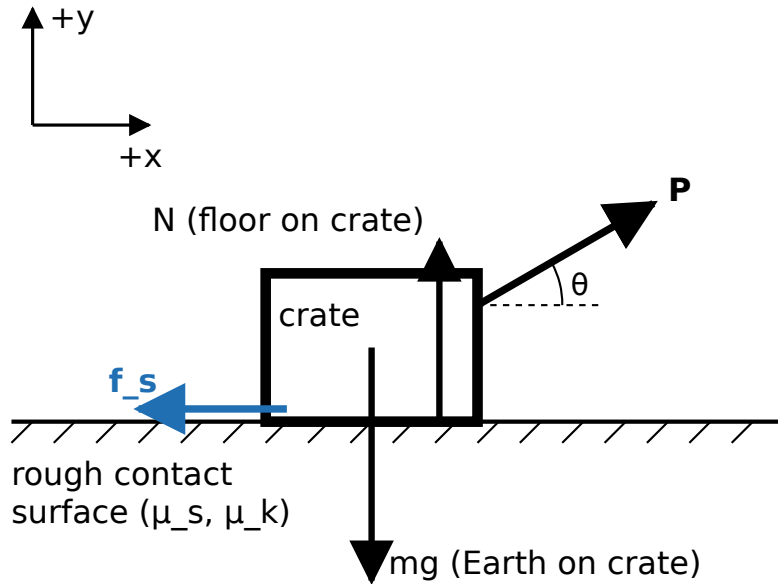
What changed

The learner must carry a force that changes both the tangential demand and the normal reaction through the static-friction threshold instead of treating $\mu_s mg$ as a fixed limit.

What stayed valid

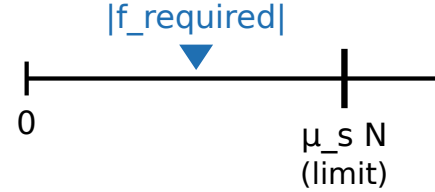
Static friction is still at most $\mu_s N$ and equals it only at impending slip; what changes is that the push also presses the block into the floor, so N grows with P .

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



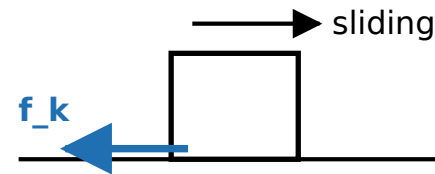
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

At impending slip, $N = mg + P \sin(\theta)$ and $P \cos(\theta) = \mu_s N$, so $P_{\max} = \mu_s m g / (\cos(\theta) - \mu_s \sin(\theta))$.

1. The downward component of P increases the normal reaction, so the vertical balance gives $N = mg + P \sin(\theta)$.
2. For a static state, friction must oppose the horizontal push; the maximum available value is reached only at impending slip.
3. At the threshold, $P \cos(\theta) = \mu_s [mg + P \sin(\theta)]$.
4. Rearranging gives $P[\cos(\theta) - \mu_s \sin(\theta)] = \mu_s mg$ and hence the stated $P_{\max}$.

Rubric

- Computes N from the vertical force balance including the downward component of P .
- Uses static-friction equality only at impending slip.
- Derives the threshold expression with the force-angle term retained in both horizontal and vertical components.

Independent check

Setting $\theta = 0$ reduces the result to $P_{\max} = \mu_s mg$, the ordinary horizontal-push threshold.

Repair

Revisit: Build the chosen-body FBD and determine the normal reaction N from the force balance or acceleration perpendicular to the contact; do not assume $N = mg$.

Lineage check

Static friction is still at most $\mu_s N$ and equals it only at impending slip; what changes is that the push also presses the block into the floor, so N grows with P .

The learner must carry a force that changes both the tangential demand and the normal reaction through the static-friction threshold instead of treating $\mu_s mg$ as a fixed limit.

A block of mass m rests on a block of mass M on a smooth floor. A horizontal force F is applied to the upper block m . The coefficient of static friction between the blocks is μ_s . Derive the largest F for which the blocks move together without slipping.

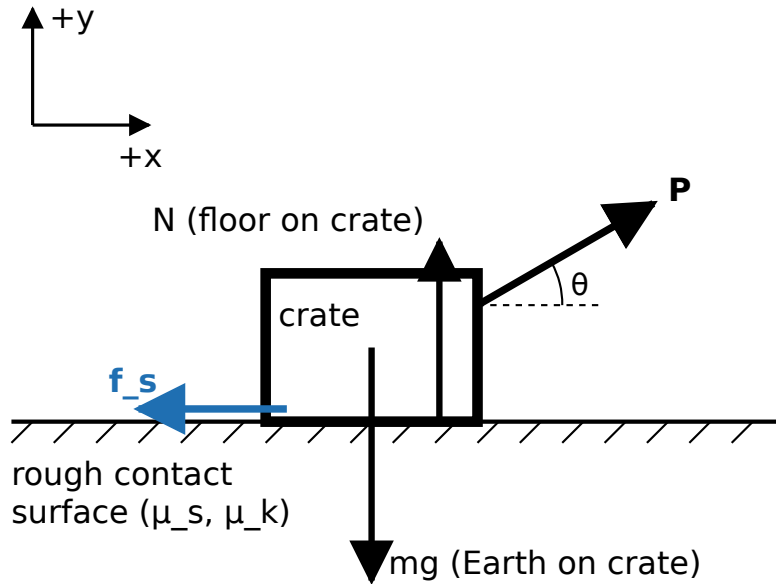
Builds on

- [A 5 kg block is already sliding to the right on a horizontal floor. A constant 18 N...](#)

Conditions

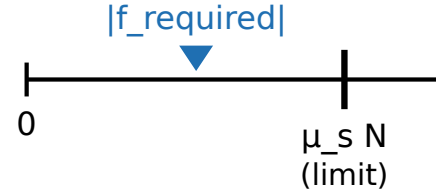
- The lower block has no other horizontal contact force from the floor.
- The contact between the two blocks remains horizontal.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



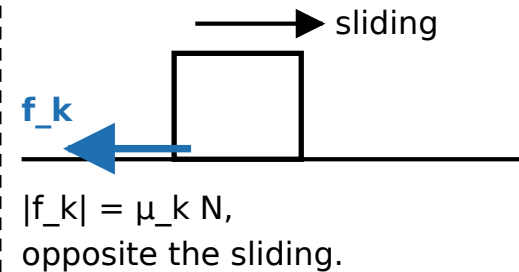
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$: Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

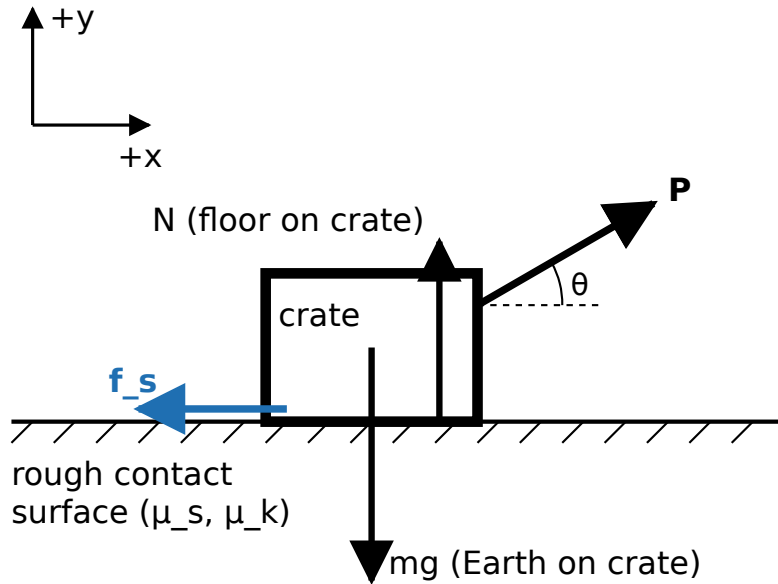
What changed

The learner must rebuild the no-slip force account after the applied force moves to the other body, rather than copying the friction equation from the earlier stacked-block threshold item.

What stayed valid

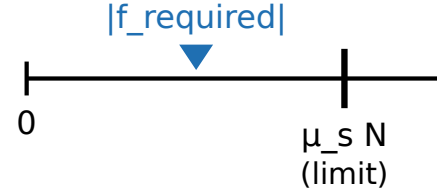
The whole system still sets the shared acceleration, and the friction one block needs is still tested against $\mu_s N$; what changes is which block needs the friction.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



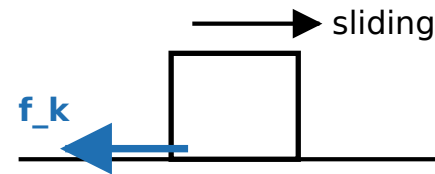
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

If the blocks move together, $a = F/(m+M)$. The lower block needs friction $f_{\text{required}} = Ma$. With $N = mg$ at the interface, no slip requires $MF/(m+M) \leq \mu_s m g$, so $F_{\text{max}} = \mu_s m g (m+M)/M$.

1. Under the no-slip assumption, treat both blocks as one system to obtain the common acceleration $a = F/(m+M)$.
2. Now isolate the lower block M . Its only horizontal force is the friction from the upper block, so $f_{\text{required}} = Ma = MF/(m+M)$.
3. The interface normal is $N = mg$ because the upper block has no vertical acceleration or other vertical force.
4. Static contact is feasible only while $f_{\text{required}} \leq \mu_s N$; equality gives the threshold F_{max} .

Rubric

- Uses one common acceleration only under the provisional no-slip constraint.
- Isolates the lower block to identify the required friction as Ma .
- Compares the required friction with $\mu_s mg$ and solves the resulting threshold.

Independent check

The friction needed to accelerate the lower block is the quantity tested against the static limit; using $f = \mu_s N$ for every smaller F would erase the dependence on F .

Repair

[Revisit: Assume no relative sliding provisionally and solve the tangential Newton-II equation for the static friction value \$f_{\text{required}}\$ that would make that motion possible.](#)

Lineage check

The whole system still sets the shared acceleration, and the friction one block needs is still tested against $\mu_s N$; what changes is which block needs the friction.

The learner must rebuild the no-slip force account after the applied force moves to the other body, rather than copying the friction equation from the earlier stacked-block threshold item.

Two blocks A and B are initially touching on a smooth horizontal floor, with A to the left of B. A horizontal force F is suddenly applied to A toward the left, away from B. A proposed solution treats $A+B$ as one connected system and writes $a=F/(m_A+m_B)$. Decide whether that model is valid immediately after the force is applied. If it is not, state the two accelerations.

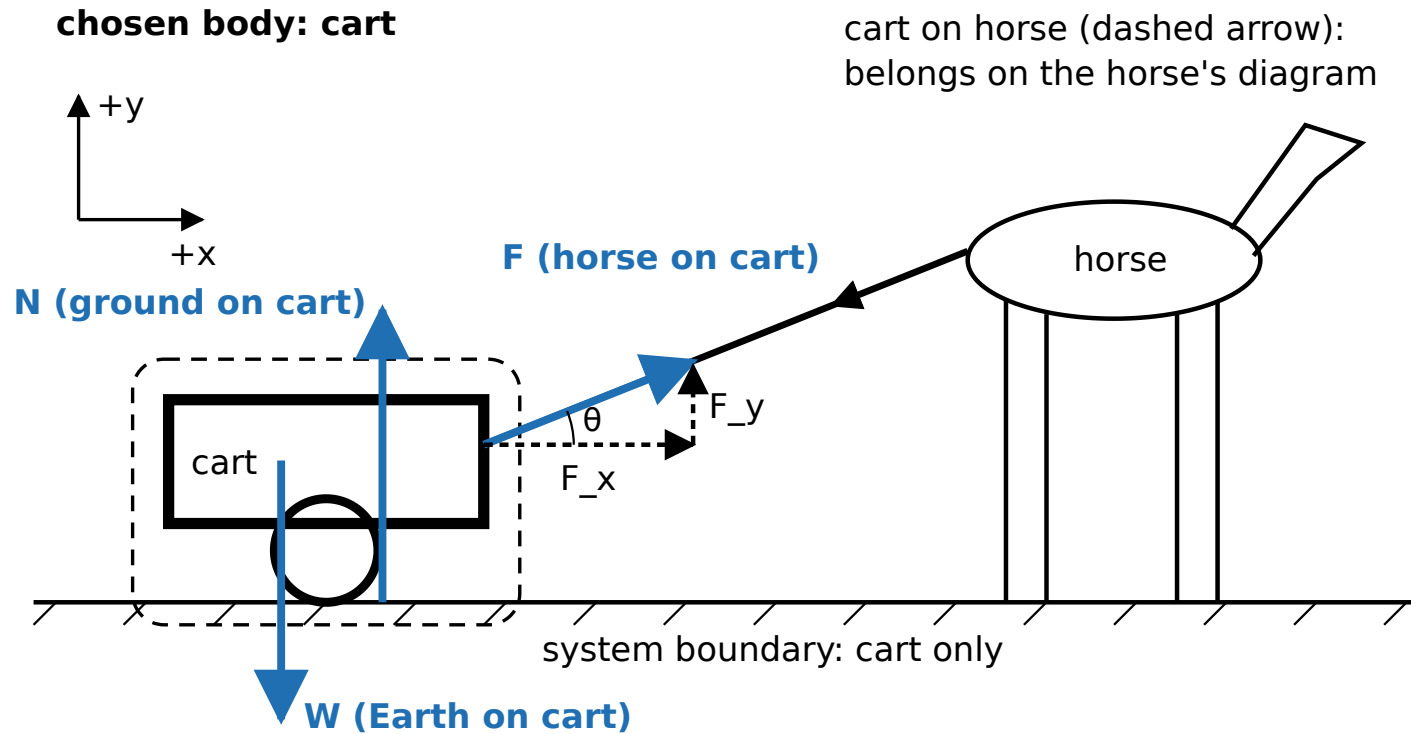
Builds on

- [Blocks A and B are side by side on a smooth horizontal floor, with A to the left of B...](#)

Conditions

- There is no glue, string, or other connector.
- The contact is an ordinary normal interaction that cannot pull.

- Take rightward as positive.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

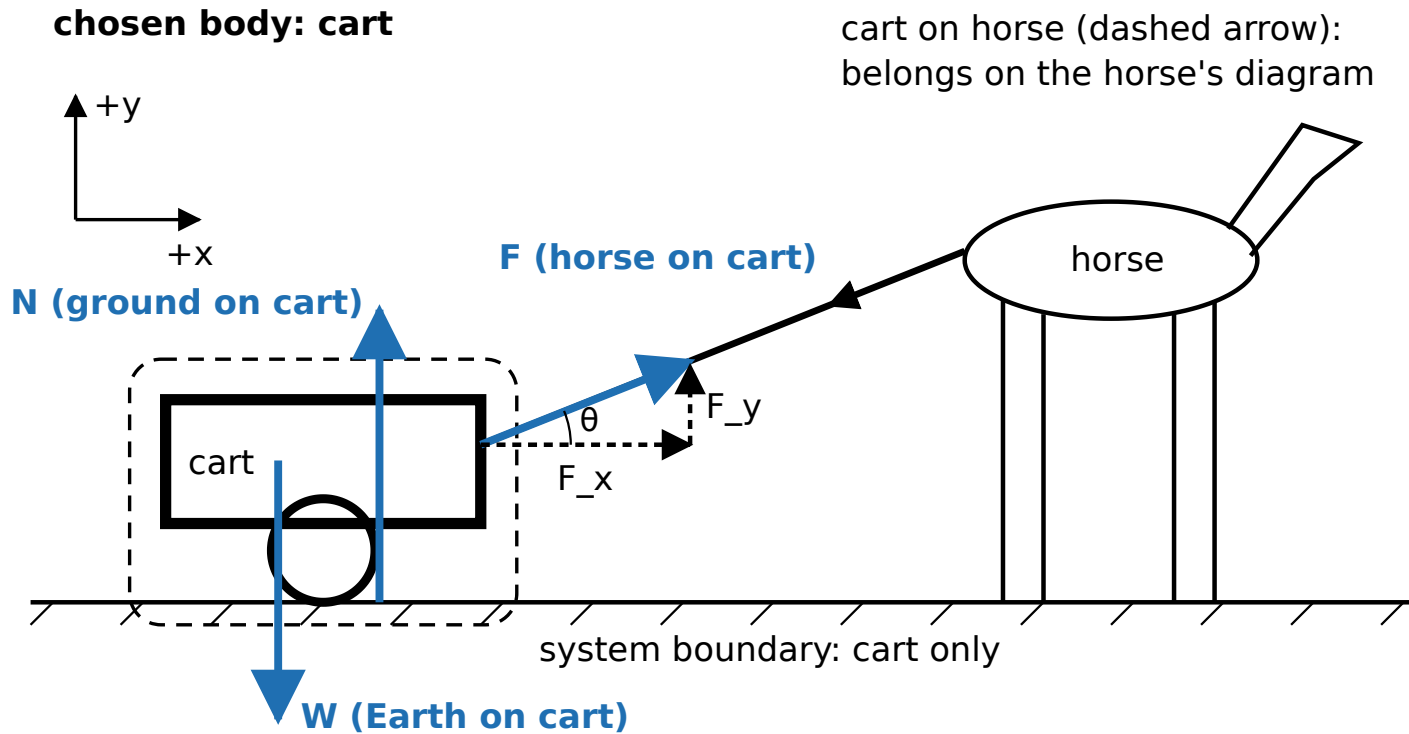
Review and solution

What changed

The learner must decide that the common-acceleration connected-system model has ceased to apply, rather than being told that contact is maintained.

What stayed valid

A common acceleration still needs an active constraint, and a contact force still can only push; what changes is that the constraint has actually been lost.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

The common-acceleration model is invalid because the force makes A separate from B and an ordinary normal contact cannot pull them together. Immediately after contact is lost, $a_A = -F/m_A$ and $a_B = 0$.

1. The earlier common-acceleration rule applies only while an active condition fixes the blocks' separation.
2. Here F drives A away from B, so maintaining contact would require a tensile contact force on B, which an ordinary normal interaction cannot supply.
3. Once the contact force becomes zero, B has no horizontal force and therefore zero horizontal acceleration.
4. A has the applied leftward force F , so with rightward positive its acceleration is $-F/m_A$.

Rubric

- Rejects common acceleration because the stated contact cannot enforce fixed separation under the applied force.
- Sets the post-separation contact force to zero.
- Obtains $a_A = -F/m_A$ and $a_B = 0$ in the declared sign convention.

Independent check

The rejected value $F/(m_A+m_B)$ would require B to accelerate left even though no horizontal interaction remains on B.

Repair

[Revisit: Identify the connection/contact condition and state exactly which relative separation is fixed.](#)

Lineage check

A common acceleration still needs an active constraint, and a contact force still can only push; what changes is that the constraint has actually been lost.

The learner must decide that the common-acceleration connected-system model has ceased to apply, rather than being told that contact is maintained.

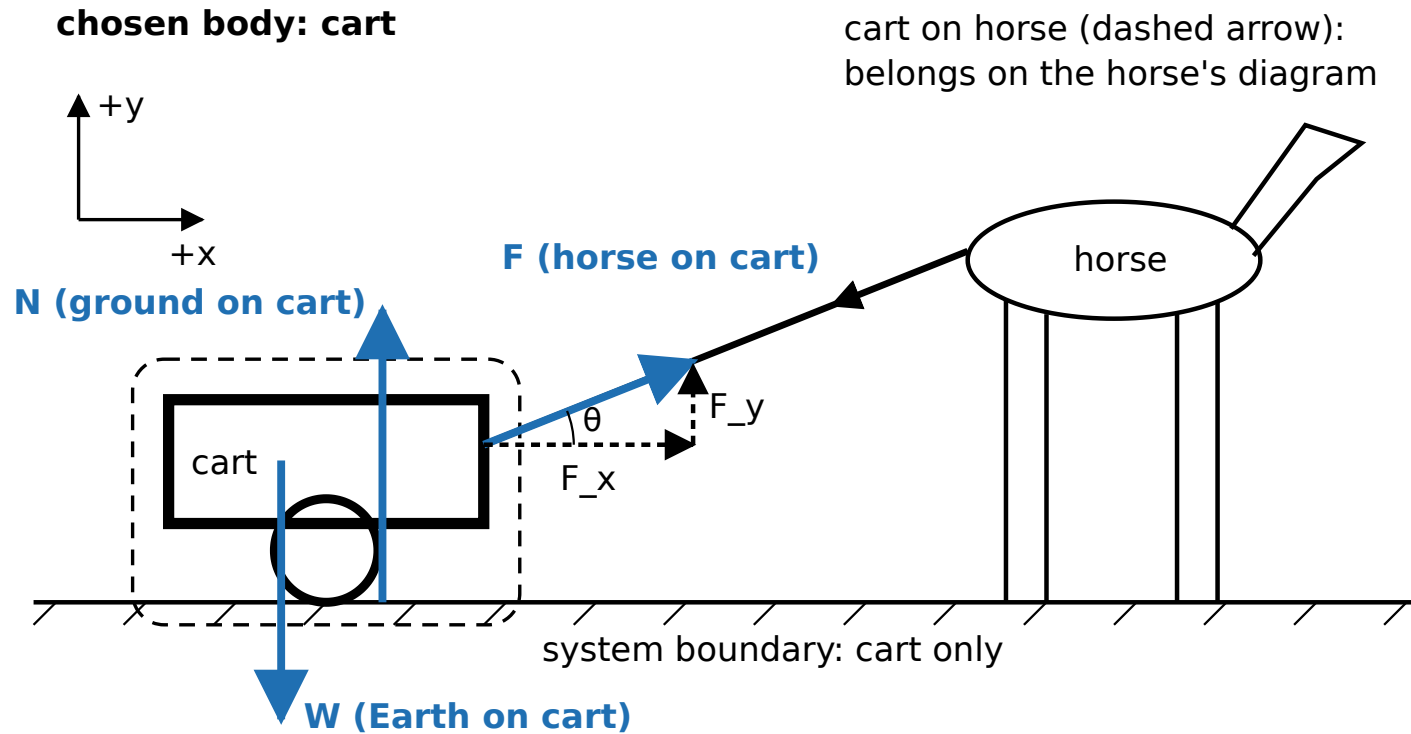
Two masses remain attached to one taut inextensible string that does not slip on a fixed pulley. The pulley is now stated to have appreciable rotational inertia, and no rotational-dynamics relation is supplied. A student claims that both $T_A = T_B$ and $a_A + a_B = 0$ must either survive together or fail together. Decide which statement can still be obtained from the bounded material, and which statement is no longer licensed.

Builds on

- [Masses \$m_A\$ and \$m_B\$, with \$m_B > m_A\$, hang from the two ends of one taut massless...](#)

Conditions

- Coordinates y_A and y_B are measured positive away from the fixed pulley along the two straight string segments.
- The string remains taut and inextensible and does not slip.
- Do not invent a torque or rotational-inertia equation.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

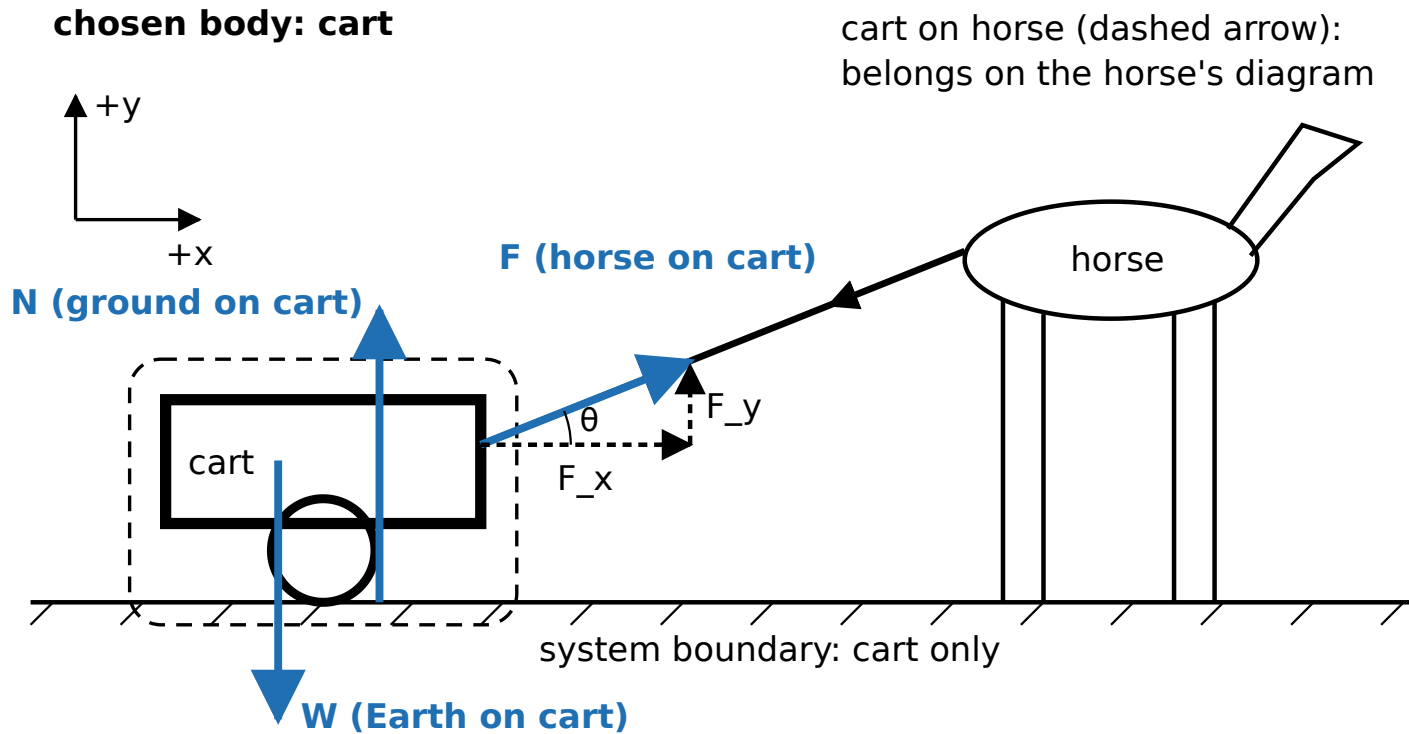
Review and solution

What changed

The learner must preserve the kinematic length constraint while rejecting the equal-tension model after only the force-transmission assumption changes.

What stayed valid

The fixed string length still gives $a_A + a_B = 0$ in away-from-pulley coordinates; what changes is that the pulley is no longer ideal, so equal tension is no longer licensed.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

The kinematic relation $a_A + a_B = 0$ still follows from the fixed total string length. Equal tension $T_A = T_B$ is no longer guaranteed by the taught ideal redirection model once appreciable pulley rotational inertia is introduced. A quantitative tension difference is outside this packet.

1. The signed acceleration relation is derived from the inextensible string's fixed length and the two changing straight segments.
2. That geometric argument does not use equality of tension magnitudes.
3. The one-common-T rule was licensed by the ideal massless/frictionless redirection model.
4. Introducing appreciable pulley rotational inertia breaks that force-transmission idealization, so the packet must refuse equal tension without supplying a replacement dynamics model.

Rubric

- Retains $a_A + a_B = 0$ from the fixed-length geometry.
- Refuses to assert $T_A = T_B$ after the ideal redirection assumption is broken.
- Does not manufacture a numerical or symbolic nonideal-pulley tension relation.

Independent check

If the pulley is restored to the massless ideal model, both claims are licensed again, but for different reasons: geometry for the acceleration relation and force transmission for equal tension.

Repair

[Revisit: Keep the kinematic string-length claim separate from the force-transmission claim: derive the acceleration relation from inextensibility and fixed geometry, and use equal tension only when the independent ideal-string/redirection assumptions license it.](#)

Lineage check

The fixed string length still gives $a_A + a_B = 0$ in away-from-pulley coordinates; what changes is that the pulley is no longer ideal, so equal tension is no longer licensed.

The learner must preserve the kinematic length constraint while rejecting the equal-tension model after only the force-transmission assumption changes.

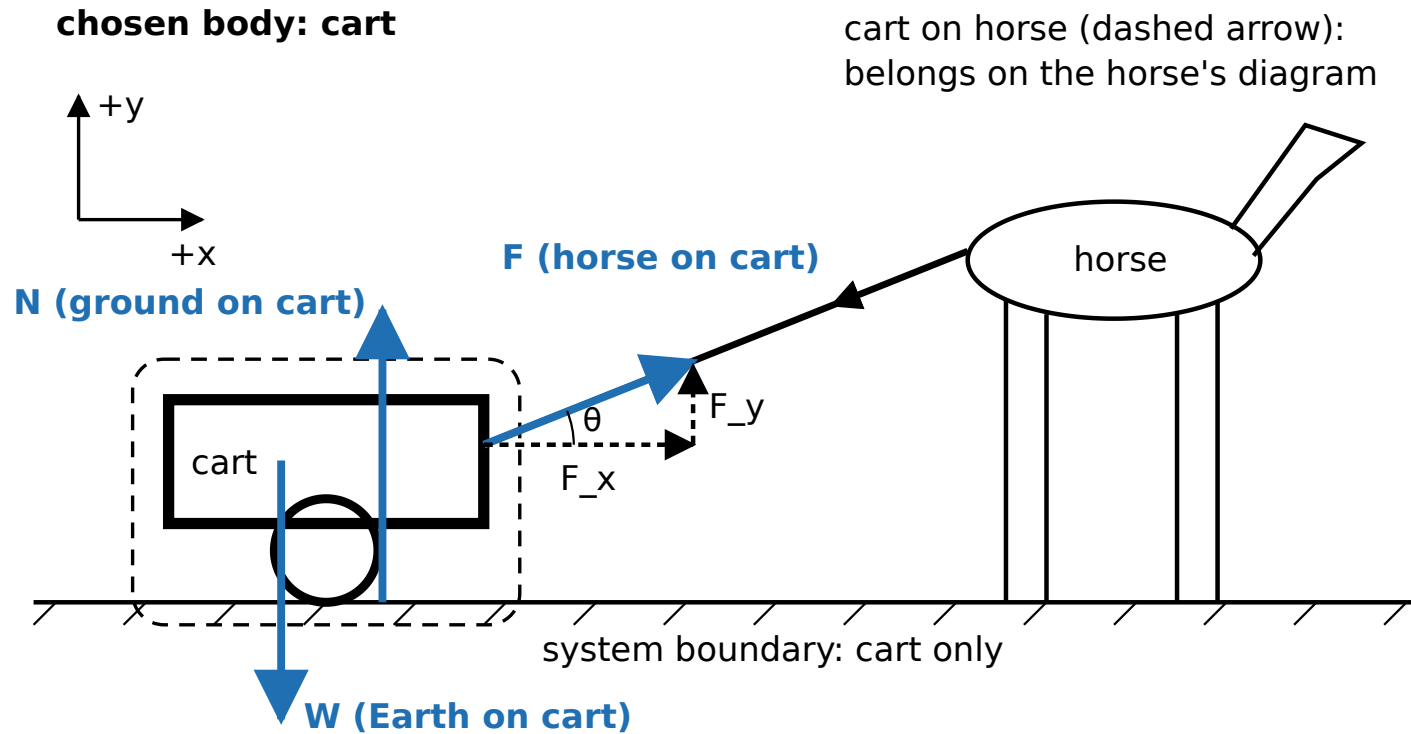
The same smooth fixed-incline problem is solved by two students. Student A uses axes parallel/perpendicular to the plane. Student B keeps horizontal/vertical laboratory axes. Decide whether both coordinate choices are physically valid, and explain why Student A's choice usually makes the contact constraint and acceleration easier to express.

Builds on

- [A block of mass \$m\$ is released from rest on a smooth fixed plane inclined at angle...](#)

Conditions

- Both students use one consistent inertial frame.
- The plane is fixed and smooth.
- No numerical angle is required.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

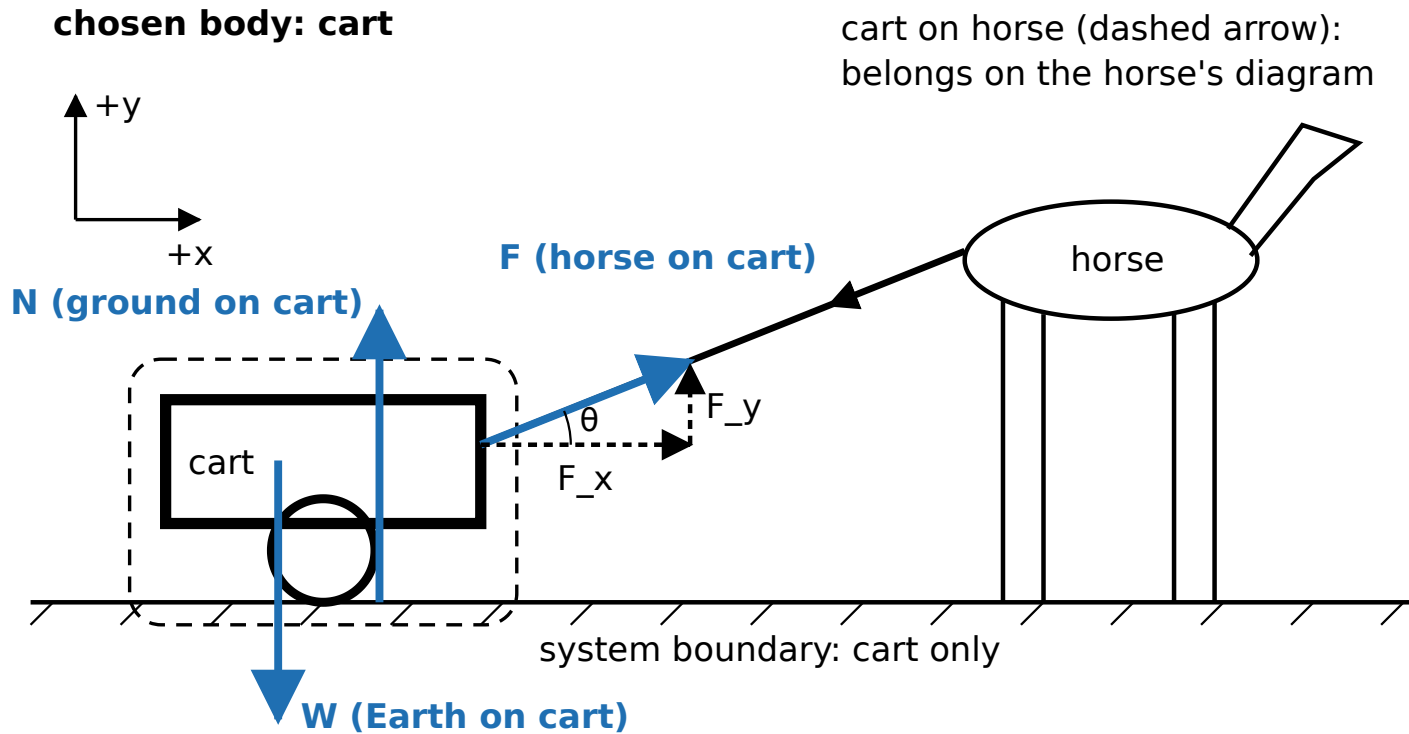
Review and solution

What changed

The learner must translate the same fixed-incline dynamics between contact-aligned and laboratory axes without treating one representation as different physics.

What stayed valid

The forces, weight and normal reaction, and the block's physical acceleration down the slope stay the same; only the axes used to describe them change.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

Both coordinate systems are valid if used consistently. Plane-aligned axes are usually simpler because maintained contact becomes $a_{\text{perp}}=0$ and the unknown acceleration has only a parallel component; horizontal/vertical axes require projecting both the normal force and the constrained acceleration.

1. Changing axes does not change the physical forces or motion.
2. In plane-aligned axes, the normal reaction lies on the perpendicular axis and the tangential motion lies on the parallel axis.
3. The contact condition is therefore stated directly as $a_{\text{perp}}=0$.
4. In horizontal/vertical axes the same normal and acceleration must be represented by components, which is valid but algebraically less direct.

Rubric

- Accepts both coordinate systems as valid representations.
- Identifies $a_{\text{perp}}=0$ as the simple contact statement in plane-aligned axes.
- States that horizontal/vertical axes require extra projection rather than different physics.

Independent check

A correct solution in either coordinate system must predict the same physical acceleration vector and normal reaction.

Repair

Revisit: Choose axes that make the active constraint easy to express; for a fixed inclined contact, use parallel/perpendicular axes when helpful, resolve the physical forces into those axes, and write separate signed Newton-II equations.

Lineage check

The forces, weight and normal reaction, and the block's physical acceleration down the slope stay the same; only the axes used to describe them change.

The learner must translate the same fixed-incline dynamics between contact-aligned and laboratory axes without treating one representation as different physics.

A block is at rest on a rough incline. A controllable force P acts up the plane. In one trial P is slightly smaller than $mg \sin(\theta)$; in another it is slightly larger than $mg \sin(\theta)$, and in both trials the available static friction is sufficient to prevent motion. Determine the static-friction direction in each trial without assuming that friction always points uphill.

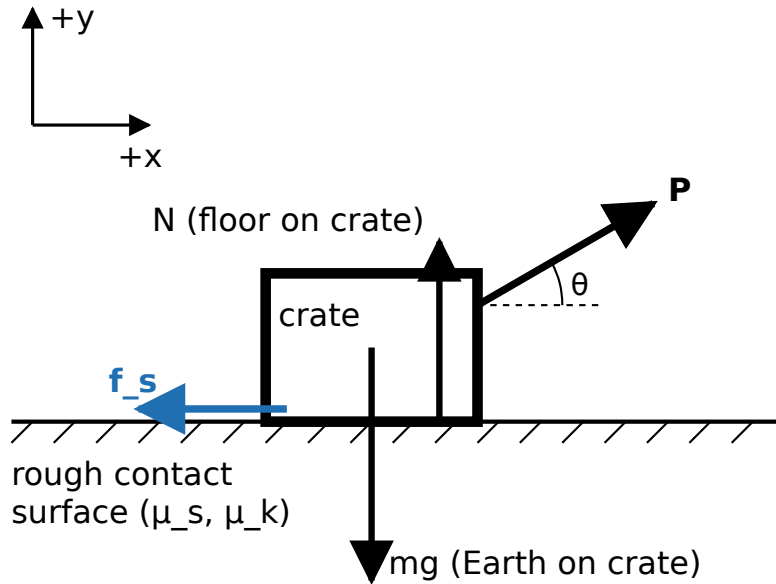
Builds on

- [A block rests on a rough fixed plane inclined at angle \$\theta\$. The coefficient of...](#)

Conditions

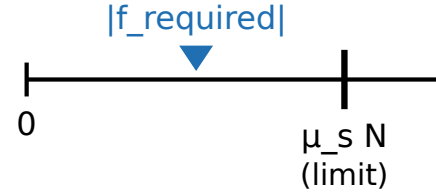
- The plane is fixed.
- The block remains at rest in both stated trials.
- P acts exactly parallel to the plane.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



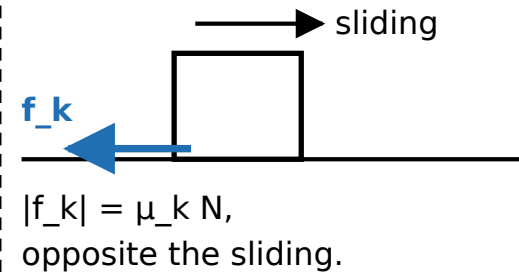
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$: Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

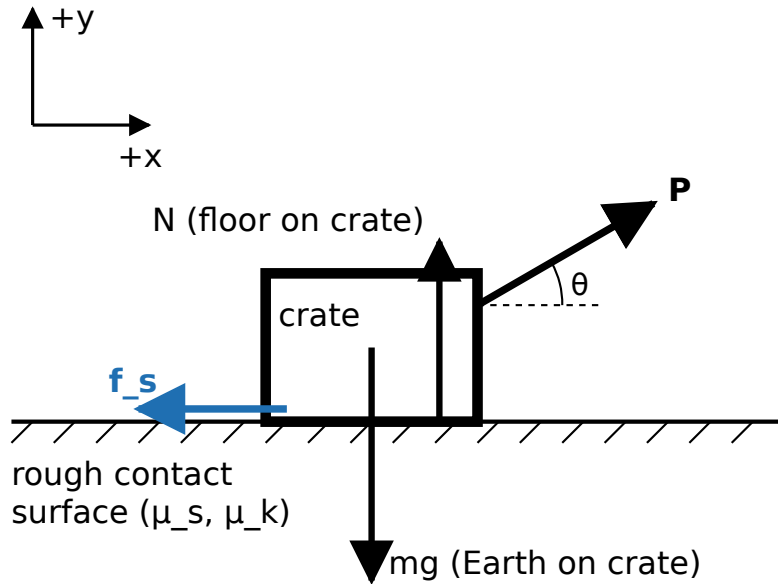
What changed

The learner must choose the static-friction direction from the changed slip tendency, including a zero-friction boundary, instead of carrying over the familiar downhill-gravity case.

What stayed valid

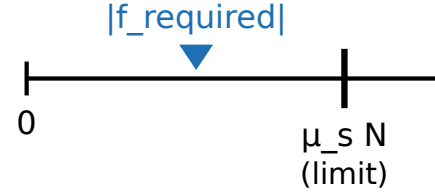
Friction still opposes the slip tendency the block would have without it; what changes is that a push along the slope can cancel that tendency or reverse it.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



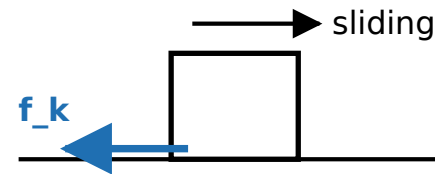
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

When $P < mg \sin(\theta)$, the no-friction tendency is down the plane, so static friction acts up the plane.

When $P > mg \sin(\theta)$, the no-friction tendency is up the plane, so static friction acts down the plane.

At $P = mg \sin(\theta)$, the required static friction is zero.

1. Friction direction follows the relative slip tendency that would exist without friction.
2. Compare the applied uphill force P with the downhill gravity component $mg \sin(\theta)$.
3. If gravity's component is larger, friction must act uphill to maintain rest.
4. If P is larger, friction must act downhill.
5. If the two non-friction tangential forces balance exactly, no tangential friction is required.

Rubric

- Uses the no-friction slip tendency rather than the object's current zero velocity.
- Assigns uphill friction for $P < mg \sin(\theta)$ and downhill friction for $P > mg \sin(\theta)$.
- Allows zero friction at $P = mg \sin(\theta)$.

Independent check

The physical direction changes continuously through a zero-friction case rather than remaining permanently uphill.

Repair

Revisit: Imagine the relative sliding or tendency to slide if friction were absent, then point friction opposite that relative tendency.

Lineage check

Friction still opposes the slip tendency the block would have without it; what changes is that a push along the slope can cancel that tendency or reverse it.

The learner must choose the static-friction direction from the changed slip tendency, including a zero-friction boundary, instead of carrying over the familiar downhill-gravity case.

A block of mass m is held at rest on a rough fixed plane that rises to the right at angle θ . A horizontal force P pushes to the right. The coefficient of static friction is μ_s . Derive the condition on P and θ for the block to be able to remain at rest.

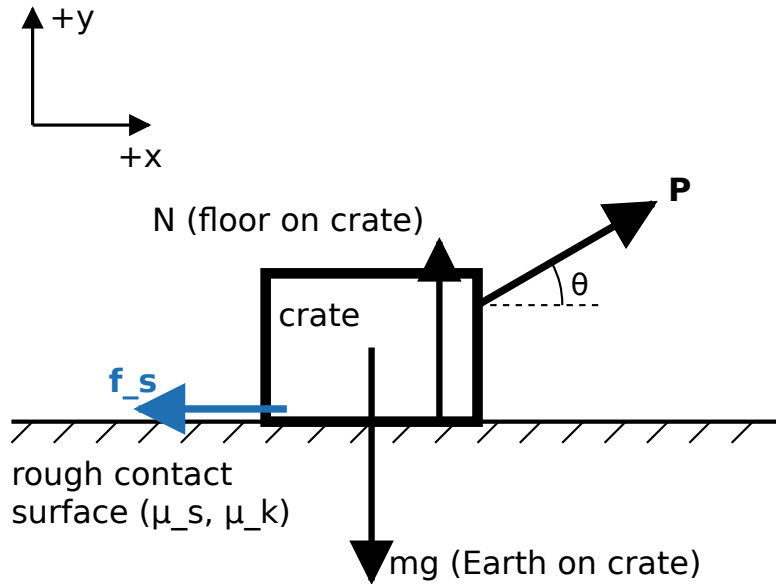
Builds on

- [A block of mass \$m\$ is on a smooth fixed plane that rises to the right at angle \$\theta\$...](#)

Conditions

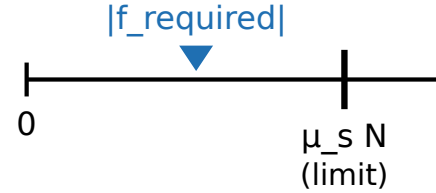
- $0 < \theta < 90$ degrees.
- The applied force is horizontal.
- Do not assume the static-friction direction in advance; use a signed required value.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



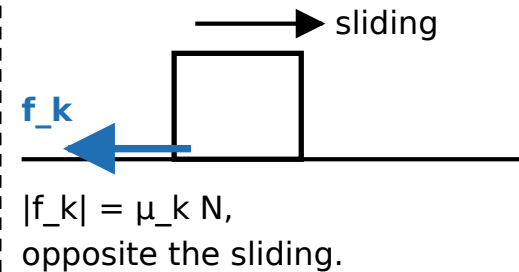
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

The learner must integrate force decomposition, altered normal reaction, signed friction demand and static feasibility in dependency order rather than copying $N = mg \cos(\theta)$ or $f = \mu_s N$.

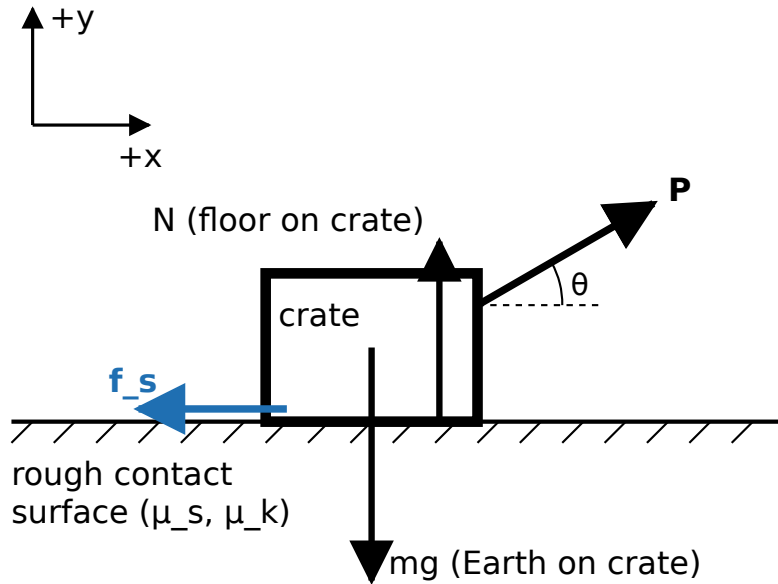
What stayed valid

The normal force from the perpendicular balance, the friction that rest requires, and the test against $\mu_s N$ all carry over; what changes is that a horizontal push alters both N and the required friction at once.

The deciding move

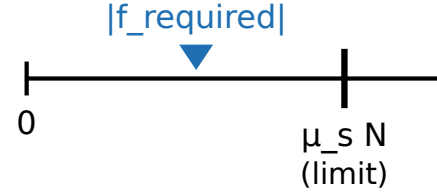
Use the sign of f_{required} to determine the required friction direction and test its magnitude against $\mu_s N$.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



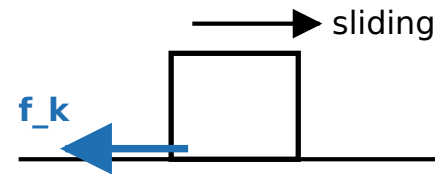
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

With uphill positive, $N = mg \cos(\theta) + P \sin(\theta)$. The static friction required by rest is $f_{\text{required}} = mg \sin(\theta) - P \cos(\theta)$. Static equilibrium is feasible exactly when $|mg \sin(\theta) - P \cos(\theta)| \leq \mu_s [mg \cos(\theta) + P \sin(\theta)]$.

1. Resolve P into $P \cos(\theta)$ uphill and $P \sin(\theta)$ into the plane.
2. The perpendicular rest condition gives $N = mg \cos(\theta) + P \sin(\theta)$.
3. The tangential rest equation $P \cos(\theta) + f_s - mg \sin(\theta) = 0$ gives the signed required friction $f_{\text{required}} = mg \sin(\theta) - P \cos(\theta)$.
4. Do not choose a friction direction separately; the sign of f_{required} supplies it.
5. The static model is feasible when the magnitude of that required value does not exceed $\mu_s N$.

Rubric

- Includes the inward perpendicular component $P \sin(\theta)$ when finding N .
- Solves a signed f_{required} from the tangential equilibrium equation.
- Tests $|f_{\text{required}}|$ against $\mu_s N$ instead of setting friction equal to its limit immediately.

Independent check

At $P=0$ the condition reduces to $\tan(\theta) \leq \mu_s$; if $P \cos(\theta) = mg \sin(\theta)$, the required friction is zero.

Repair

Revisit: Build the chosen-body FBD and determine the normal reaction N from the force balance or acceleration perpendicular to the contact; do not assume $N = mg$.

Lineage check

The normal force from the perpendicular balance, the friction that rest requires, and the test against $\mu_s N$ all carry over; what changes is that a horizontal push alters both N and the required friction at once.

The learner must integrate force decomposition, altered normal reaction, signed friction demand and static feasibility in dependency order rather than copying $N = mg \cos(\theta)$ or $f = \mu_s N$.

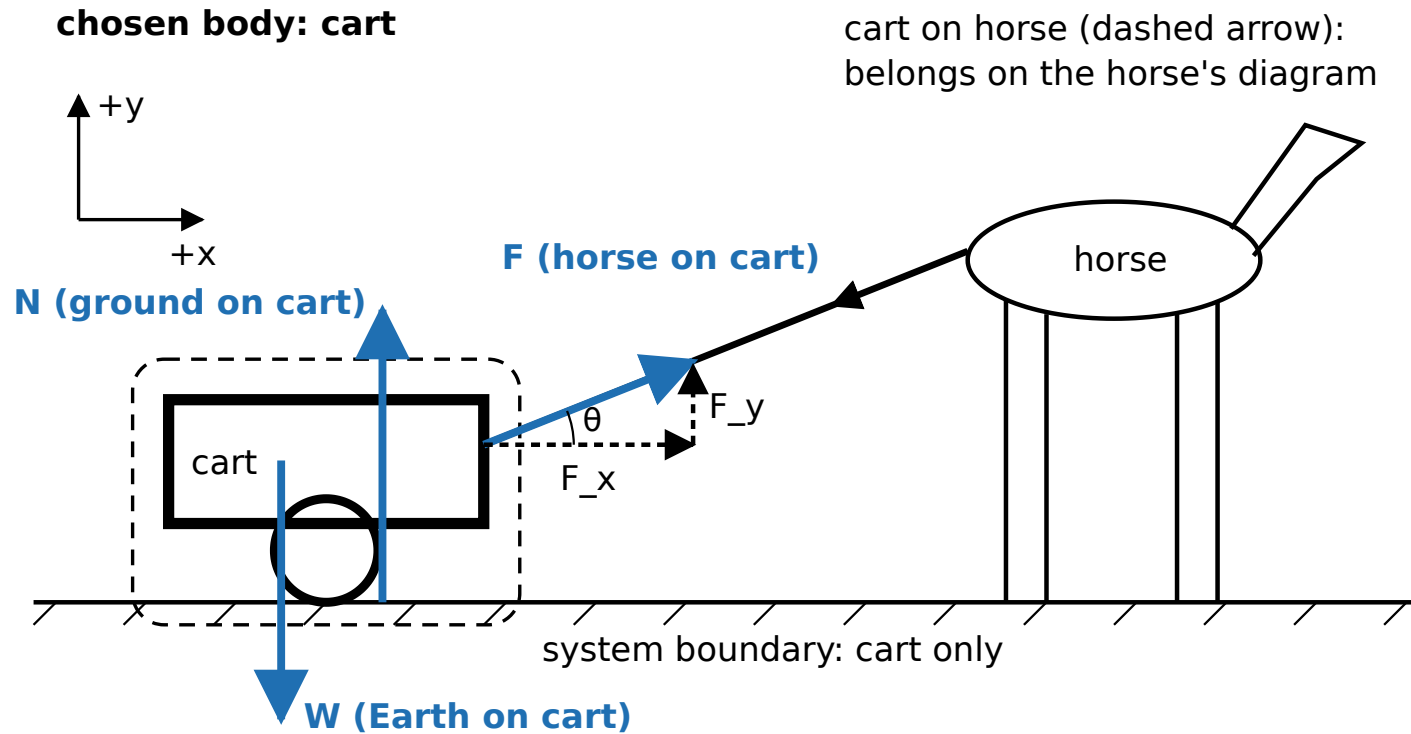
A proposed plane-aligned solution writes the tangential force sum as $mg + mg \sin(\theta)$ and the perpendicular force sum as $N - mg - mg \cos(\theta)$. The solver says the component terms were added because the weight was resolved. Identify the modelling error and write the correct role of the weight in the two component equations.

Builds on

- [A block rests on a smooth fixed incline. A student draws three downward-related force...](#)

Conditions

- The block is on a smooth fixed incline.
- Only gravity and the normal reaction act.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Commit to a model or representation and your first relation before opening support.

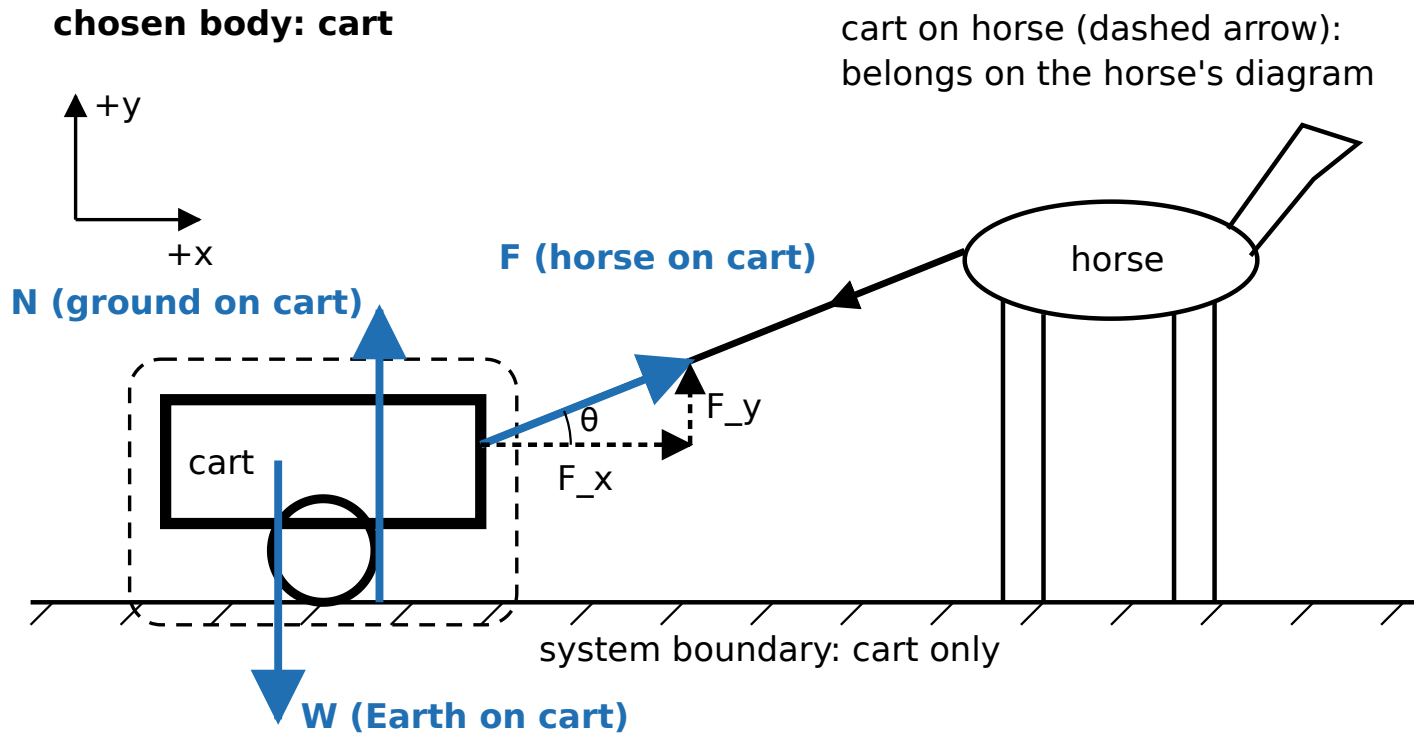
Review and solution

What changed

The learner must recognize the same force/component double-counting error after it moves from a drawn FBD into algebraic component equations.

What stayed valid

Weight is still one force described by two components, and the normal force is still the only other interaction; what changes is that the double count appears in equations rather than on a drawing.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Answer

The solution double-counts the same weight vector. In plane-aligned equations, gravity contributes $mg \sin(\theta)$ down the plane and $mg \cos(\theta)$ into the plane; the unresolved mg is not added again. Thus the tangential gravity term is $mg \sin(\theta)$, and the perpendicular equation uses $N - mg \cos(\theta) = 0$ for maintained contact.

1. Weight mg is one physical force, not three.
2. Resolving it replaces its representation in the chosen component equations.
3. The original vector magnitude mg is not itself a signed scalar component along either tilted axis.
4. Use the two appropriate components in the two equations and keep the normal as the separate contact force.

Rubric

- Identifies double-counting of gravity as the error.
- Uses $mg \sin(\theta)$ only in the tangential gravity contribution.
- Uses $mg \cos(\theta)$ only in the perpendicular gravity contribution and N separately.

Independent check

The corrected equations account for exactly two physical interactions: Earth-on-block gravity and plane-on-block normal contact.

Repair

Revisit: When a physical force is resolved into components for chosen axes, keep the body-force inventory unchanged: use the components as a representation of that one force in the equations, not as additional interaction forces alongside the original vector.

Lineage check

Weight is still one force described by two components, and the normal force is still the only other interaction; what changes is that the double count appears in equations rather than on a drawing.

The learner must recognize the same force/component double-counting error after it moves from a drawn FBD into algebraic component equations.

A block starts from rest and travels the same distance s down the same incline of angle θ in two trials. On a smooth surface the time is t_0 . On a rough surface the block slides throughout and the time is $t > t_0$. Assuming kinetic friction coefficient μ_k is constant, derive μ_k in terms of θ , t_0 and t .

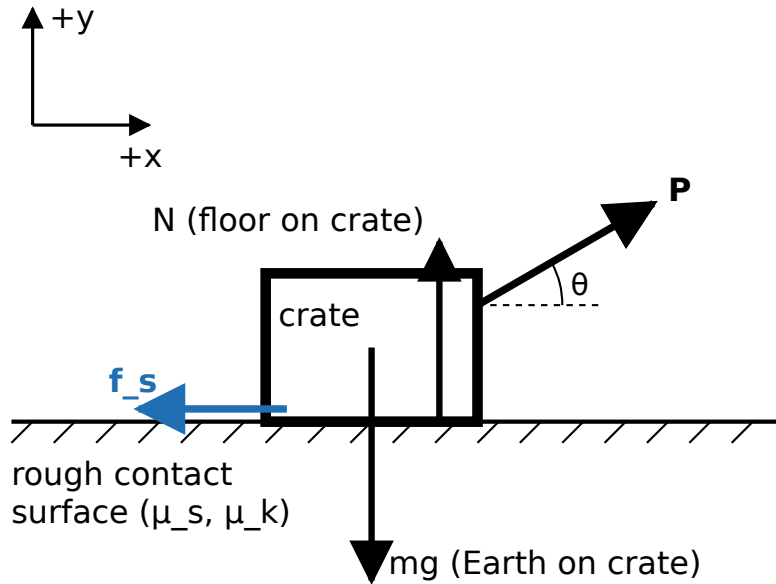
Builds on

- [A block is already sliding down a rough fixed plane inclined at angle \$\theta\$. The...](#)

Conditions

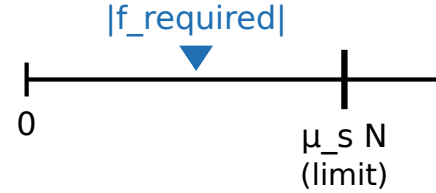
- Both trials start from rest at the same point.
- The incline angle and travel distance are unchanged.
- The rough-surface trial has established downward sliding with constant μ_k .

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



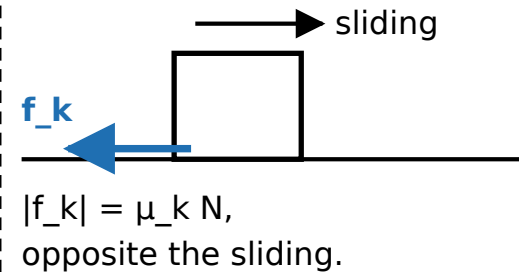
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Commit to a model or representation and your first relation before opening support.

Review and solution

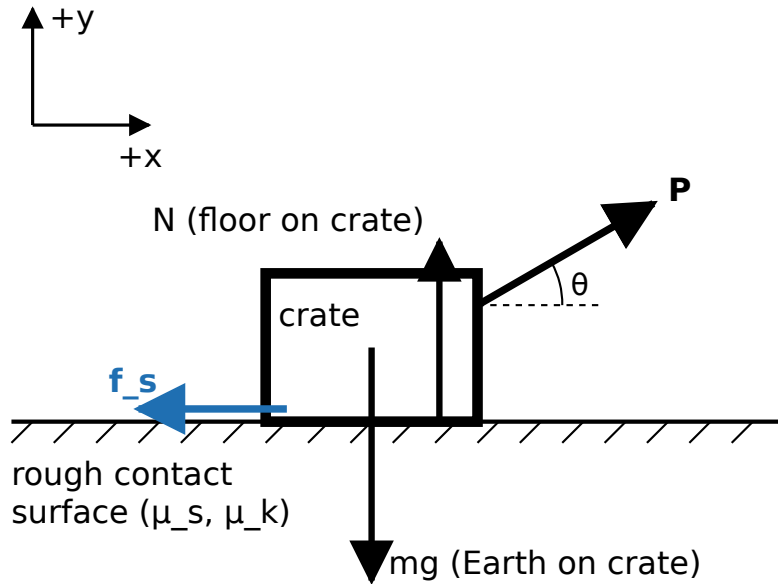
What changed

The learner must combine the established rough-incline dynamics with an independent timing representation to infer the friction coefficient, rather than being asked directly for acceleration.

What stayed valid

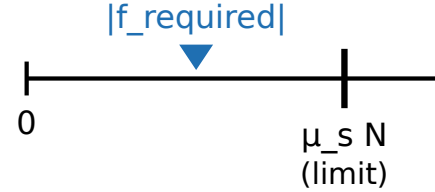
The rough-slope acceleration $g(\sin \theta - \mu_k \cos \theta)$ and the smooth-slope $g \sin \theta$ still hold; what changes is that μ_k must be inferred from travel times instead of being used to find an acceleration.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



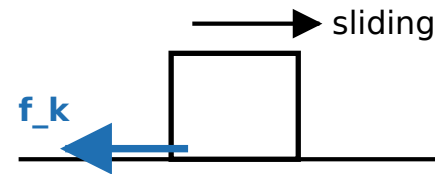
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Answer

Smooth acceleration is $a_0 = g \sin(\theta)$, rough acceleration is $a = g[\sin(\theta) - \mu_k \cos(\theta)]$. Since $s = (1/2)at^2$ from rest, $a/a_0 = t_0^2/t^2$. Therefore $\mu_k = \tan(\theta)[1 - t_0^2/t^2]$.

1. The smooth trial gives $a_0 = g \sin(\theta)$.
2. The rough sliding trial gives $a = g[\sin(\theta) - \mu_k \cos(\theta)]$.
3. Equal distance from rest gives $s = (1/2)a_0 t_0^2 = (1/2)a t^2$, hence $a/a_0 = t_0^2/t^2$.
4. Substitute the two incline accelerations and solve for μ_k .
5. This yields $\mu_k = \tan(\theta)[1 - t_0^2/t^2]$.

Rubric

- Builds the smooth and rough incline accelerations from the corresponding force models.
- Uses equal-distance constant-acceleration kinematics to relate a/a_0 to t_0^2/t^2 .
- Solves to $\mu_k = \tan(\theta)[1 - t_0^2/t^2]$ and checks its limiting sign.

Independent check

Because $t > t_0$, the bracket is positive. If t approaches t_0 , the inferred μ_k approaches zero.

Repair

Revisit: Use $|f_s| = \mu_s N$ only at impending slip; if relative sliding is already established, use $|f_k| = \mu_k N$ with direction opposing the relative sliding.

Lineage check

The rough-slope acceleration $g(\sin \theta - \mu_k \cos \theta)$ and the smooth-slope $g \sin \theta$ still hold; what changes is that μ_k must be inferred from travel times instead of being used to find an acceleration.

The learner must combine the established rough-incline dynamics with an independent timing representation to infer the friction coefficient, rather than being asked directly for acceleration.